

Re-visiting the Political Economy of the Rise and Fall of the Unidad Popular: Towards a Global Political Economy Approach

Diego Polanco¹

Department of Economics, University of Massachusetts Amherst

September 2026

Abstract

This chapter investigates how central bank reserve solvency mediates the inflationary transmission of domestic money creation during the Chilean Unidad Popular (1970–1973). To evaluate the contested breakdown of Salvador Allende’s socialist government, I assemble a high-frequency monthly panel (1960–1980) and a secular annual dataset (1920–2010) to test competing monetarist, Austrian, and structural-dependency hypotheses. Using forensic Granger causality tests and a 3-regime Threshold Vector Autoregression (TVAR) conditioned on the Central Bank Solvency Ratio ($\text{SolvR}_t \equiv e_t \cdot IR_t/M_{1,t}$), I demonstrate that monetary expansion operated as an endogenous accommodation valve responding defensively to external bottlenecks and distributive conflict, refuting orthodox exogeneity axioms. Monetization is inflationary only when external solvency is depleted: the monthly pass-through to inflation is statistically zero in the solvent Hedge regime ($\text{SolvR}_t > 0.21$), +0.54% in the intermediate Speculative regime, and +4.54% once external blockade pushes the economy into a Ponzi regime ($\text{SolvR}_t \leq 0.14$). Audited balance-of-payments ledgers confirm that the Nixon financial embargo and multilateral credit freeze inflicted an immediate capital account contraction of −\$294.0 million in 1971 (98.1% of the reserve drain), unmasking acute structural import bottlenecks. Macroeconomic collapse reflected external balance-sheet exhaustion under international currency hierarchy, operating through foreign credit freezes and imported intermediate shortages.

Keywords: Unidad Popular, Balance of Payments, Financial Subordination, Solvency Ratio, Endogenous Money, Threshold Vector Autoregression, International Currency Hierarchy.

JEL Codes: B51, C32, E58, F33, N16.

Contents

1	Introduction	4
2	Contested Approaches to the Political Economy of the Rise and Fall of the Unidad Popular	6
2.1	Classical Dependency Theory: Interrupted Programme	7
2.2	The Rise of the Chicago Boys and the Neo-liberal School	8
2.3	The Neo-Structuralist School: Selective Sanitization	9
2.4	Historical Political Economy: Methods Under Inherited Macro Theory	11
2.5	The Revival of Dependency Theory as a <i>Research Programme</i>	12
2.6	Epistemological Critique: The Illusion of Causality, Narrative License, and Col- lider Traps	13
3	World Money, Financial Subordination, and the Peripheral Central Bank Balance Sheet	14
3.1	World Money, the Universal Equivalent, and the Monetary Expression of Value .	14
3.2	The Bretton Woods Architecture and Peripheral Balance-of-Payments Constraints	15
3.3	The Peripheral Central Bank Balance Sheet: From Reserve Adequacy to Leverage Dynamics	15
3.4	Money Endogeneity vs. Exogeneity and the Peripheral Paradox	16
3.5	The Peripheral Central Bank Balance Sheet under International Financial Sub- ordination	16
3.6	The Four Structural Dimensions of the Unidad Popular Crisis	18
4	Data Architecture, Archival Sources, and Variable Construction	18
4.1	Archival Provenance and Multi-Frequency Research Design	18
4.2	Variable Construction, Accounting Identities, and Boundary Conditions	19
5	Historical Background, Stylized Facts, and Empirical Results	20
5.1	Historical Background: The Rise of the Unidad Popular	20
5.2	Stylized Facts of Peripheral Accumulation, Distribution, and External Constraints	20
5.3	Empirical Exercise: <i>Lakatosian</i> Granger Causality Tournament	21
5.3.1	Econometric Framework: Toda–Yamamoto Levels Augmentation	21
5.3.2	Univariate Stationarity and Integration Pre-testing Battery ($d_{\max} = 1$) . .	21
5.3.3	Sequential Tournament Architecture and Hypothesis Testing Rules	21
5.3.4	Empirical Findings	21
5.3.5	Residual Diagnostics and Dynamic Stability	21
5.4	Empirical Exercise: Non-Linear Dynamics under Peripheral Central Bank Sol- vency. A Threshold VAR Approach	21
5.4.1	The Threshold VAR Specification and Structural Identification	21
5.4.2	Endogenous Threshold Estimation and Residual Bootstrap Linearity Test- ing	21
5.4.3	Regime Parameter Estimates and System Stability Forensics	21
5.4.4	Dynamic Generalized Impulse Response Functions (GIRFs)	21
5.4.5	Formal Multiplier Difference Hypothesis Tests	21
6	Political Economy Discussion, Global Implications, and Concluding Synthe-	

sis	21
6.1 The Political Economy of Peripheral Solvency and Subordinated Money	21
6.2 The Unidad Popular in the Global Structural Crisis of the 1970s	22
6.3 Policy Implications for Contemporary Peripheral Sovereignty	22
6.4 Concluding Synthesis: Toward a Progressive Research Programme	23

Appendices	30
Appendix A Formalization of Solvency Ratio and Balance-of-Payments Accounting	30
A.1 The Solvency Ratio in Nominal Terms	30
A.2 Current and Capital Account Decomposition	31
A.3 Derivation of the Discrete Solvency Law of Motion	31
A.4 Analytical Convexity and Non-Linear Reserve Exhaustion	32
A.5 Constraint-Binding Conditions and Structural Minskian Regimes	33
Appendix B Monthly TVAR Impulse Response Atlas (1960–1980)	33
Appendix C Integration Order Diagnostics and Unit Root Battery	40
Appendix D Archival Data Provenance, Variable Dictionary, and Splicing Protocols	43
D.1 Monthly High-Frequency Variable Ledger	43
D.2 Monetary Reform Conversions and Splicing Mechanics	43
Appendix E Non-Parametric Causal Discovery (Runge et al., 2019) as Structural Robustness for the Monthly Threshold VAR Architecture	45
E.1 The PCMCI Algorithmic Protocol	45
E.2 Empirical Directed Acyclic Graph (DAG) Findings	45
E.3 Topological Robustness Mapping to the Threshold VAR Architecture	45
Appendix F Extended Archival Ledgers of Inquilinaje, Agrarian Labor Petitions, and Union Density (1932–1975)	48
F.1 Annual Archival Ledger of Agricultural Labor Petitions (1932–1950)	48
F.2 Strike Typology and Legal Status (1960–1975)	49
F.3 Trade Union Affiliation and Union Density Ledger (1932–1975)	49
F.4 Urban Land Seizures in Santiago and Nationwide (1967–1971)	49
F.5 Productive Capital Stock, Profit Rate, and Net Accumulation Framework (1960–1975)	50

1 Introduction

1.1 In the history of economic thought, progress rarely unfolds as a linear, cumulative ascent toward universal truth (Roncaglia, 2006). As Alessandro Roncaglia demonstrated in his epistemological defense of the discipline, theoretical advance is governed by a *competitive view*: an enduring contest among rival paradigms rooted in fundamentally distinct conceptualizations, analytical categories, and visions of the world. For over half a century, the historiography of Chile’s Unidad Popular (UP, 1970–1973) under Salvador Allende has been interpreted through an uncritical cumulative lens. Mainstream accounts project neoclassical factor-clearing assumptions and closed-system monetary axioms onto peripheral reality, treating the Chilean breakdown as a settled cautionary tale of populist fiscal profligacy. Yet beneath this orthodox consensus lies an unexamined failure of conceptualization: treating the peripheral state as a closed monetary system, blind to the structural boundaries imposed by the world market, international currency hierarchies, and balance-of-payments constraints (Vasudevan, 2009; Basu, 2022).

1.2 This chapter investigates how central bank reserve solvency mediates the inflationary transmission of domestic money creation during peripheral socialist transformation. If money creation mechanically drove inflation, why did base money expand by over 100% in 1971 while monthly inflation remained quiescent at 1.5–2.0% and annual inflation fell from 35% to 22%? The answer lies in balance-sheet solvency: in 1971, the Central Bank possessed \$400M in inherited foreign reserves and industry operated with ample idle capacity ($\mu \approx 82\%$), absorbing liquidity through real transactions demand. Inflation pass-through did not exist in 1971—it emerged only in mid-1972 once reserves fell below \$30M and the international credit blockade severed revolving commercial trade lines. Broad money growth functioned as an endogenous accommodation valve through which the central bank mediated an imperial balance-of-payments blockade and intense distributive conflict, contesting the monetarist thesis of Edwards (2024) and the Austrian interpretation of Espinosa and Véjar (2025). I formalize the central bank balance sheet through the Solvency Ratio—the ratio of convertible foreign reserve assets to domestic monetary liabilities ($\text{SolvR}_t \equiv \frac{e_t \cdot IR_t}{M_{1,t}}$)—identifying it as the state-dependent threshold governing monetary transmission. My formal threshold tests decisively reject linear and two-regime models in favor of a three-regime architecture extending the Financial Instability Hypothesis of Minsky (1986) to open-economy central banking under international currency hierarchies (Foley, 2003; Chamorro-Futínico, 2025). In the solvent Hedge regime ($\text{SolvR}_t > 0.21$), foreign reserve backing absorbs domestic liquidity, holding monetary pass-through to prices at zero. In the intermediate Speculative regime ($0.14 < \text{SolvR}_t \leq 0.21$), reserve erosion strains credibility and generates moderate price pass-through (+0.54%). When external credit embargoes and private capital flight breach the lower threshold ($\text{SolvR}_t \leq 0.14$), the central bank enters a Forced Ponzi regime, where defensive liquidity accommodation to maintain productive solvency passes directly into hyperinflation (+4.54%) and currency flight.

1.3 Allende’s thousand days unfolded within the tectonic shifts of the global structural crisis of capitalism of the early 1970s. The unilateral suspension of dollar-gold convertibility by the United States on August 15, 1971 dismantled the Bretton Woods system, detonating world commodity stagflation and intensifying imperial conflict over peripheral natural resources (Harmer, 2011; Arboleda, 2023). Simultaneously, the secular post-war decline of the corporate profit rate across the advanced capitalist core (Weisskopf, 1979; Bowles et al., 1983) narrowed the geopolitical room for peripheral developmentalism. The collapse of the Chilean socialist transition represented the structural collision of domestic redistributive aspirations with external financial subordination under an unstable international monetary order.

1.4 Mainstream and neoclassical syntheses characterize the Allende administration through the paradigm of “macroeconomic populism” (Dornbusch and Edwards, 1990; Larraín and Meller, 1991). In this interpretation, redistributive governments inevitably trace an unforced four-phase

cycle: initial reactivation through idle capacity, the emergence of bottleneck shortages, unsustainable fiscal and monetary financing, and final economic collapse. By treating the central bank as an unconstrained printing press and monetary emission as an unmediated political choice, this literature commits two deep epistemological errors. First, it succumbs to what [Blanco and Matute \(2018\)](#) formulate as the *illusion of causality*—a cognitive heuristic that infers direct causal generation from the mere temporal coincidence of salient events (the 1970 election, 1971 wage readjustments, and 1972–1973 inflation) while ignoring non-contingent external covariates. Second, it exercises what [Isch et al. \(2026\)](#) classify as *narrative license*—the rhetorical escalation of observational and correlational snapshots into ironclad causal laws to satisfy academic demands for compelling monocausal parables. By analyzing annual aggregates in an institutional vacuum, mainstream models mistake the physical symptoms of organized distribution sabotage and external financial asphyxiation for the mechanical results of domestic monetary policy.

1.5 To discriminate between these contested claims on rigorous empirical ground, I deploy an applied Lakatosian evaluation of contested scientific research programmes (MSRP) in economic thought ([Lakatos, 1978](#); [Roncaglia, 2006](#); [King and Sznajder, 2006](#); [Kvangraven, 2021](#)). This methodological framework appraises competing programmes by distinguishing their immutable theoretical axioms (*hard core*) from the auxiliary specifications of their empirical models (*protective belt*), evaluating whether empirical anomalies are absorbed through theoretically progressive extensions that predict *novel facts* or through ad hoc degenerating immunizing stratagems:

1. **The Monetarist Programme (H_1 , [Edwards, 2024](#)):** Its hard core asserts exogenous base money creation, fractional-reserve multiplier causality ($H \rightarrow M1 \rightarrow P$), and long-run monetary neutrality in a closed system. Faced with the empirical anomaly of non-inflationary 100% monetary growth in 1971, its protective belt degenerates into ad hoc narrative stratagems that treat the \$294.0M external financing collapse, the U.S. Eximbank blockade, and domestic distribution sabotage as incidental background noise.
2. **The Austrian Business-Cycle Programme (H_2 , [Espinosa and Véjar, 2025](#)):** Its hard core posits that artificially suppressed interest rates and credit expansion distort intertemporal price signals, inducing a malinvestment boom in higher-order, roundabout capital goods ($\Delta K \gg 0$), whose inevitable coordination failure precipitates systemic breakdown.
3. **The Structural-Dependency Programme (H_3 , [Foley, 2003](#); [Vasudevan, 2009](#); [Chamorro-Futinico, 2025](#)):** Advanced in this chapter, its hard core roots peripheral macroeconomics in international currency hierarchies, balance-of-payments dominance, conflict inflation, and credit money endogeneity (Loans $\rightarrow$ Deposits $\rightarrow$ Reserves). Its protective belt extends Minsky’s financial fragility taxonomy to the central bank balance sheet ($\text{SolvR}_t \equiv e_t \cdot IR_t / M_{1,t}$), achieving a progressive problemshift that successfully predicts the novel empirical fact of state-dependent threshold non-linearity: domestic liquidity expansion generates zero consumer price inflation under solvent reserve buffers ($\text{SolvR}_t > 0.21$), jumping into explosive pass-through only when external asphyxiation breaches the Forced Ponzi threshold ($\text{SolvR}_t \leq 0.14$).

1.6 This chapter deepens the theoretical foundations established in Chapter 2. Whereas Chapter 2 identified the external boundary of accumulation through an annual balance-of-payments threshold ($\gamma = 0.425$) that throttles mechanization under unbalanced growth, this chapter uncovers the dynamic balance-sheet mechanism through which that external boundary operates. To avoid temporal aggregation bias, I coordinate two empirical tiers: a secular annual panel (1920–2010, $T = 91$) and a high-frequency monthly archival panel (1960–1980, $N = 250$ continuous months). At the annual frequency, H_2 fails decisively: gross fixed investment in machinery

and equipment contracted (-2.3%) during the 1971 expansion while structural capacity utilization surged to an all-time peak of $\mu_{1971} = 0.9844$, confirming Keynesian-Kaleckian reserve reactivation rather than Austrian capital lengthening. At the monthly frequency, the 3-regime TVAR and event studies capture the opening of a relational trade account scissors operating across two distinct domains. On the physical and real purchasing power domain, Prebisch-ECLAC capacity to import (M_t^{cap}) collapsed to 27.7 following the August 1971 Nixon Shock and copper price decline, blowing open the Real Structural Imbalance ($\Theta_t \equiv M_t/M_t^{\text{cap}} \times 100$, operationalizing the Latin American structuralist concept of *desequilibrio estructural externo*) to an unsustainable peak of 283.4 in May 1973. This real physical imbalance constitutes the certified structural shock driving industrial bottlenecks and inflation in our DAG causal discovery. Concurrently, on the nominal and exchange-rate domain, the Relative Price Disparity Index ($\Omega_t \approx 1/\text{RER}_t$) exploded to 190.6 under the official parity peg ($e_t = 0.0122$). Because imperial credit embargoes enforced hard quantity rationing on physical imports in foreign exchange, Ω_t diverged from effective import volumes, recording the nominal price disparity preserved by the state's defense of the social wage.

1.7 The remainder of this chapter is structured as follows. Section 2 surveys the contested traditions in Chilean political economy and formalizes the epistemological critique of orthodox causality. Section 3 develops the theoretical framework of World Money, monetary endogeneity, and the open-economy transposition of Minskyan solvency regimes to the peripheral central bank. Section 5 grounds the inquiry in the historical political economy of the pre-1970 accumulation regime, details the multi-frequency data architecture, specifies the model restrictions, and implements the econometric evaluation across the four-stage empirical strategy. Section 6 discusses the political economy of peripheral solvency, situates the UP within the global structural crisis of the 1970s, and outlines lessons for contemporary monetary sovereignty.

2 Contested Approaches to the Political Economy of the Rise and Fall of the Unidad Popular

2.1 No single framework commands consensus on the political economy of the rise and fall of Chile's Unidad Popular (UP). Rival explanatory traditions have organized the field instead, and they emerged under sharply unequal historical conditions. Transnational networks linking the Pontificia Universidad Católica de Chile to the University of Chicago consolidated the neoliberal school, which then acquired state power through the dictatorship that followed the 1973 coup. Neo-structuralism emerged under that same dictatorship as a technocratic opposition and later became the preferred economic language of the post-dictatorship order. Historical Political Economy entered much later, bringing powerful tools of causal identification to historical questions, but entering a disciplinary environment already shaped by earlier victories and exclusions. Dependency theory occupies a different position altogether. It was the intellectual world from which the UP programme itself emerged, and the coup shattered it institutionally before it could resolve the theoretical problems it had opened.

2.2 These traditions did not simply coexist. They succeeded one another through conflict. Each identified part of the problem, and none integrated external transmission, distributive conflict, and macroeconomic dynamics into a single explanatory sequence. This section critically evaluates these traditions. It reconstructs the historical conditions under which each approach became authoritative, clarifies the mechanisms each privileged, and shows which questions each rendered visible and which it displaced, sanitized, or foreclosed. It begins with classical dependency theory, which posed the problem the chapter as a whole seeks to resolve more sharply than any approach that followed.

2.1 Classical Dependency Theory: Interrupted Programme

2.3 Dependency theory did not approach the Unidad Popular from the outside. It helped constitute the intellectual horizon within which the UP became thinkable. Its formation in Chile brought together two lineages that converged, institutionally and theoretically, in Santiago during the long sixties. From one side came the Monthly Review tradition, beginning with Baran (1957)'s analysis of economic surplus under monopoly capitalism and radicalized by Frank (1967) into the development-of-underdevelopment thesis: underdevelopment was not a residual stage but an active product of incorporation into the world capitalist system. From the other came the structuralist school built within the headquarters of the Economic Commission for Latin America (ECLAC), led by Raúl Prebisch, which had already framed Latin American development through centre-periphery relations, external bottlenecks, and the historical asymmetries of international trade.

2.4 At the Centro de Estudios Socioeconómicos (CESO) of the Universidad de Chile, the two lineages fused into a common research programme. Theotônio dos Santos, Vania Bambirra, André Gunder Frank, Orlando Caputo, and Roberto Pizarro, in close dialogue with Ruy Mauro Marini in Concepción, transformed dependency from a general critique of peripheral capitalism into an attempt to specify the historical forms linking regional subordination, accumulation, and political strategy in Latin America (Dos Santos, 1970; Bambirra, 1971; Caputo and Pizarro, 1970; Kay, 2010; Cárdenas et al., 2022).

2.5 The programme's analytical wager was clear. Peripheral capitalism followed a distinct developmental dynamic structured by external constraints, a bourgeoisie incapable of autonomous development, and patterns of accumulation that continually reproduced subordination. Within this tradition, socialist transformation was an objective developmental necessity grounded in the structural limits of peripheral capital accumulation. Yet the tradition never achieved theoretical unity. Marini's account of superexploitation pushed the analysis toward a world-market logic in which the systematic compression of labour conditioned domestic reproduction; Cardoso and Faletto insisted instead that dependency took historically specific forms depending on the configuration of domestic class alliances and the relative autonomy of the state (Marini, 1973; Cardoso and Faletto, 1979). The tension was decisive. At issue was whether domestic political mediation constituted an independent object of analysis or followed largely from world-market imperatives. The tradition posed the question with great force and never settled it.

2.6 That unresolved tension did not prevent dependency theory from producing the richest contemporaneous political-economy account of the UP. De Vylder (1976)'s *Allende's Chile* remains a seminal account seldom referenced in Chilean debates: a participant-observer reconstruction of the UP as a process shaped by the interaction of internal class dynamics and external constraint. Around it emerged a broad empirical corpus tracking the same intertwined problem from complementary angles—distributional conflict across successive governments, worker and peasant radicalization from below, the external credit squeeze and spare-parts embargo, the monetary and balance-of-payments mechanisms through which external pressure became domestic crisis, and the unresolved strategic problem of the state under conditions of dual power (Stallings and Zimbalist, 1975; Stallings, 1978; Espinosa and Zimbalist, 1978; Kay, 1978; Petras and Morley, 1975; Griffith-Jones, 1981; Sideri, 1979; Sweezy and Magdoff, 1974; Zavaleta Mercado, 1974; Winn, 1986). The tradition's strength lay precisely here. It did not treat external intervention and domestic conflict as competing explanations; it approached them as interacting dimensions of the same historical process.

2.7 That achievement also marked the limit dependency theory could not overcome in its classical form. The tradition posed the integrated problem correctly: how external subordination, domestic class struggle, and macroeconomic instability became mutually conditioning dynamics

in peripheral capitalism. It lacked a unified theory of the state and a formal specification of the transmission mechanisms linking external shocks to domestic distributive and macroeconomic outcomes. Its categories were historically powerful, but the internal architecture required to operationalize them remained unfinished. The weakness lay in the modelling rather than the diagnosis. The tradition identified the decisive mechanisms and could not yet represent them in an integrated way.

2.8 The defeat of dependency theory was twofold. Politically, the coup dismantled the institutions, networks, and research centres in which the programme had been elaborated. Intellectually, the interruption arrived before the tradition could work through its own internal tensions. No superior explanatory framework refuted it. It broke under conditions in which theoretical incompleteness coincided with the collapse of Chile's liberal democracy, the failure of democratic socialism to resolve the crisis under organized domestic and international sabotage, and the installation of a subsequent civic-military dictatorship. Subsequent approaches entered a field already reorganized by force, not reason. The next school to become authoritative in Chilean political economy was not the one that best resolved the problem dependency had identified, but the one that arrived coordinated with the counterrevolutionary reconstruction of the state.

2.2 The Rise of the Chicago Boys and the Neo-liberal School

2.9 The neoliberal school did not emerge from within the intellectual world of the Unidad Popular as one interpretation among others. It entered by a different route: as a transnationally trained programme of economic reconstruction that acquired authority through the counterrevolutionary reorganization of the Chilean state. Its genealogy is now well established. The 1956 agreement between the Pontificia Universidad Católica de Chile and the University of Chicago opened the institutional channel through which Milton Friedman, Arnold Harberger, and Gary Becker trained a cohort of Chilean economists in monetarism and public finance orthodoxy. By the late 1960s that network had outgrown its academic form. It had become a politically connected project, tied to the Chilean right and increasingly oriented toward a systemic restructuring of Chilean capitalism (Fischer, 2009; Valdés, 1995; Clark, 2017).

2.10 That programme found its canonical expression in *El Ladrillo*². The document matters because it codified in advance the economic architecture on which the post-UP order would be constructed. It operated as a *counterrevolutionary vanguard design*: a strategic programme prepared before the coup to direct the reconstruction of Chilean capitalism once the popular government was overthrown.

2.11 Its diagnostic identified a familiar set of problems: persistent inflation, external disequilibria, agricultural stagnation, low growth, and distortions associated with state-led development. Some of these observations captured real structural tensions within Chilean capitalism, and they deserve acknowledgement on their own terms. The argument that prolonged relative-price distortions hindered export diversification and contributed to balance-of-payments stress cannot simply be dismissed, nor can the claim that developmentalist policy generated enduring macroeconomic tensions (De Castro, 1992). The broader theoretical hypothesis holding these observations together was much weaker: that the crisis of the UP merely culminated a long secular drift toward the socialization of the economy. That claim recoded the conjuncture normatively to advance a project of market discipline. It did not explain it.

2.12 Here lay the neoliberal school's constitutive limit. It could register imbalances, but only by stripping them of their political-economy content. Distribution appeared not as class conflict but as a problem of distorted price signals. External crisis became the endogenous consequence of statist excess, never the interaction between domestic struggle and international destabilization.

² "The Brick" in English.

The October 1972 lockout, the U.S. credit squeeze, the spare-parts embargo, and the broader campaign of destabilization could not enter the account without undermining the framework's own premises. Neoliberalism did not omit these elements by accident. Its explanatory architecture required them to disappear, since admitting them would have meant abandoning the very theory that made the crisis legible as proof of the impossibility of socialist transformation.

2.13 The neoliberal interpretation thus became authoritative on conditions that were not primarily intellectual. Its rise reflected no superior integration of external transmission, distributive conflict, and macroeconomic dynamics. A programme prepared before the coup entered a field reorganized by force, and the dictatorship then installed it as the common sense of economic reconstruction. That reconstruction dismantled the institutions of political mediation and collective decision-making while relying, paradoxically, on the planning capacities of the developmental state it claimed to supersede (Clark, 2017; Bockman, 2019). The neoliberal school was less a successful solution to the explanatory problem dependency had posed in the periphery than a victorious political project presenting itself as economic science. Its enduring relevance lies precisely there: it turned one historically specific resolution of crisis into a universal lesson about the alleged impossibility of any alternative to market coordination (Espinosa, 2021).

2.3 The Neo-Structuralist School: Selective Sanitization

2.14 The neo-structuralist school was an intellectual accomplishment of the post-dictatorship order, not a simple continuation of classical structuralism. It built its institutional base under dictatorship, above all in the CIEPLAN research centres, where former structuralists and opposition economists monitored the Chicago Boys' experiment while preparing an alternative programme for democratic transition. That genealogy explains much of the school's subsequent authority. The CIEPLAN economists were more than moderates who had softened their earlier commitments under adverse conditions (Silva, 1991). They became the technocratic fraction through which the post-dictatorship centre-left could criticize orthodox neoliberalism while remaining within a framework compatible with profit-oriented, market-disciplined peripheral capitalist development. Neo-structuralism entered the field as a politically serviceable renovation suited to Chile's integration into globalized modernization (Sklair, 2000; Leiva, 2008; Maillet et al., 2016).

2.15 The decisive theoretical operation of that renovation was the abandonment of the surplus approach. Where Prebisch and the earlier ECLAC tradition had centered the appropriation and reinvestment of surplus under peripheral conditions, neo-structuralism recast the problem in the language of market failure, macroprudence, and pragmatic state intervention. The shift did not simply update structuralism. It altered its theoretical object. The external constraint remained, but increasingly as a macroeconomic restriction rather than as a political economy of surplus appropriation and dependent reproduction. Neo-structuralism reformulated a theory of historically structured subordination as a problem of policy calibration. This analytical retreat explains why Leiva (2008)'s diagnosis remains so essential: without the surplus approach, the external constraint survives only in a thinner form, legible as a balance-of-payments constraint but stripped of its class content and of the social relations that reproduce it. Neo-structuralism did not abandon empirical rigour. It abandoned the only framework in which the external constraint and distributive conflict could be theorized together as moments of the same accumulation process.

2.16 The school's account of the UP crystallized most clearly around Bitar's contribution. Originally published in 1979 and repeatedly revisited thereafter, Bitar (2013) offered the most sophisticated neo-structuralist reconstruction of the UP's crisis. Its great strength lay in refusing

a purely monetary reading of the balance of payments. Bitar sought to integrate external constraint, political opposition, and structural transformation into a single narrative, and his concept of a “viability area” remains a genuine analytical contribution: the shrinking space within which the UP programme might have advanced without precipitating macroeconomic collapse. The achievement is important and merits direct recognition. But it also established the school’s defining omission. Across multiple editions, Bitar never substantively engaged the dependency literature that had already developed around the same empirical terrain. Accounts from the classical dependency tradition such as [De Vylder \(1976\)](#) and [Stallings \(1978\)](#) were available, and Bitar never confronted them. The silence was no bibliographic accident. It marked the point at which the school displaced dependency’s central problem from legitimate post-dictatorship economic reasoning.

2.17 That displacement became hegemonic through the populist-macro paradigm. [Larraín and Meller \(1991\)](#) absorbed the UP into a four-phase sequence of initial success, emerging disequilibrium, macroeconomic deterioration, and collapse, while [Dornbusch and Edwards \(1991\)](#) generalized this logic into a regional diagnosis of “economic populism.” The construction had real analytical power. It offered a clean narrative, identifiable turning points, and policy lessons that travelled throughout the region in the aftermath of dictatorship. But that was also its function. By defining redistributive macroeconomic strategy as “macroeconomic populism,” the paradigm foreclosed the possibility that wage-led expansion might, under some conditions, be analytically sustainable rather than inherently self-defeating. [Meller \(2000\)](#) consolidated the mainstream narrative of the UP: macroeconomic irresponsibility combined with political polarization. The result was a reading self-critical in tone but exculpatory in effect, in which the programme failed because of its own excesses, not because of the interaction among external constraint, class conflict, and institutional sabotage. Sergio [Ramos \(1979\)](#) and Joseph [Ramos \(1980\)](#) capture the contrast with unusual clarity: the former treated inflation as a contradiction internal to a process of structural transformation under class struggle, whereas the latter recast crisis as a problem of disequilibrium management and stabilization failure.

2.18 Neo-structuralism actively replaced the conflict-centered structuralist problematique available from within the UP’s own intellectual production with a depoliticized substitute ([Larraín and Meller, 1991](#); [Dornbusch and Edwards, 1991](#); [Meller, 2000](#)). The school’s conceptual limitations constituted strategic omissions built directly into the terms of its post-dictatorship authority. Its primary accomplishment was to preserve elements of the structuralist external-constraint framework while converting them into the language of macroeconomic governance that would sustain two decades of centre-left hegemony. In doing so, it recast the thousand days of Allende’s government as a cautionary tale about redistributive excess. The school functioned as the progressive idiom through which post-dictatorship neoliberal hegemony could be criticized, managed, and stabilized without reopening the class-conflict categories that had made earlier structuralist and dependency traditions politically dangerous for the ruling classes ([Leiva, 2008](#)).

2.19 This assessment does not deny the analytical substance of authors such as Bitar or Meller. It specifies the price of their renovation. The inclusion of monetary, structural, and political variables in an integrated narrative was a real gain. The exclusion of class conflict as an endogenous economic force was the condition under which that gain became institutionally admissible. That is why neo-structuralism remains relevant beyond its own immediate milieu. It became the default macroeconomic memory inherited by the next generation of scholarship, including the field that would later arrive with stronger empirical tools but without a comparable theory of the underlying mechanism.

2.4 Historical Political Economy: Methods Under Inherited Macro Theory

2.20 Historical Political Economy (HPE) represents a genuine methodological advance in the study of economic history. By bringing the tools of the credibility revolution into historical inquiry, it has generated a level of empirical precision that earlier traditions rarely achieved. Its institutional consolidation around the *Journal of Historical Political Economy* and the *Oxford Handbook of Historical Political Economy* reflects a broader ambition to recover historical questions for economists without sacrificing standards of causal identification (Jenkins and Rubin, 2024; Charnysh et al., 2023). In the Chilean case, that ambition has yielded an impressive body of work on prices, wages, inequality, agrarian reform, and political violence. No serious assessment of the field can dismiss these achievements.

2.21 HPE did not create the empirical terrain on which it operates in Chile. That terrain also rests on a broader renewal of Chilean economic history, which has reconstructed long-run series on prices, trade, living standards, and distribution with considerable archival depth. The work of Mario Matus, Manuel Llorca-Jaña, Claudio Robles, and related scholars belongs to that growing economic-history field, which has made more ambitious forms of historical political economy empirically possible, rather than to HPE in the stricter disciplinary sense. Their contribution is foundational: they help constitute the quantitative and historical terrain on which later identification-driven work has gained traction. Rodríguez Weber (2018) matters especially for this chapter and dissertation, because the infrastructure built for long-run inequality series also underpins the wage-share series employed here, assembled through the work of Astorga (2023). That long-run account places the rise and fall of Allende’s government within a broader cycle of declining inequality and explicitly recognizes coercive distribution as a determinant of income shares, providing a hinge between economic history and the larger body of political-economy questions at stake.

2.22 HPE proper enters at a different level, combining historical questions with explicit identification strategies. Aldunate et al. (2022) isolate the firm-level effects of U.S. banking covert action during the Allende years, while Jaimovich and Toledo-Concha (2021) trace the long-run persistence of Mapuche territorial conflict to the uneven geography of agrarian reform and counter-reform. Girardi and Bowles (2018) should be read somewhat differently. Their event-study of the Santiago stock exchange provides a micro-level account of financial-market volatility in defense of private property, and for that reason forms a useful hinge toward dependency theory’s concern with financial subordination. These are major contributions, but they do not all do the same work. Some build the empirical archive, others identify causal effects, and a few open questions that point beyond HPE’s inherited macroeconomic frame.

2.23 That limitation appears most clearly in the work of Felipe González and his collaborators, who stand at the centre of the contemporary HPE literature on Chile (González, 2013; González and Vial, 2021; Aldunate et al., 2022; González et al., 2020; Bautista et al., 2023; Cuesta et al., 2015; González and Prem, 2026). Their contributions are methodologically exemplary: they have built new archival datasets, deployed transparent causal-identification strategies, and opened major empirical questions on agrarian reform, nationalization, covert U.S. intervention, and the institutional legacies of dictatorship. González’s research programme represents the field at its strongest. Yet it also reveals the field’s defining limit. The economy appears primarily as a terrain on which political and institutional effects can be measured, not as the constitutive terrain of class conflict, accumulation, and crisis. Even where the underlying materials point toward broader macroeconomic questions, the explanatory frame typically comes from post-dictatorship mainstream accounts rather than from within historical political economy itself. The result is a body of scholarship of very high empirical quality that displaces political economy in the classical sense into a micro-political science of institutions, interventions, and outcomes.

2.24 This boundary stems directly from the field’s disciplinary constitution. HPE approached the Chilean case after the earlier conflict among explanatory traditions had already been settled institutionally, inheriting a macroeconomic vocabulary shaped by the post-dictatorship synthesis that recoded the UP as a problem of macroeconomic populism and balance-of-payments disequilibrium. The framework developed here builds upon HPE’s empirical advances while embedding them within an explanatory programme capable of treating external transmission, distributive conflict, and macroeconomic dynamics as mutually conditioning structural processes.

2.5 The Revival of Dependency Theory as a *Research Programme*

2.25 The return to dependency theory recovers a vital research programme whose historical retreat coincided with a broader contraction of developmental horizons across Latin America. As [Palma \(2016\)](#) suggests, the demise of dependency theory and structuralism altered academic debate while impoverishing the region’s capacity to conceive development as a problem of autonomy, structural transformation, and historical agency. The analytical imperative today is to reconstruct dependency as a progressive framework capable of generating novel, falsifiable empirical explanations while reclaiming the systemic alternatives that orthodox models foreclose.

2.26 That is the significance of [Kvangraven \(2021\)](#)’s intervention. Recasting dependency theory through Lakatos’s notion of a scientific research programme, she argues that its hard core remains analytically productive: global capitalism tends to reproduce hierarchical relations between core and periphery, and those relations condition the developmental corridor within which peripheral accumulation unfolds. The move is more than an epistemological rescue operation. It is a methodological directive. It restores dependency theory as a body of scholarship capable of disagreement in auxiliary hypotheses while retaining a common research orientation. The revival asks whether the tradition can once again generate novel, empirically verified explanations of peripheral reproduction.

2.27 The programme’s contemporary reconstruction becomes sharper in [Madariaga and Palestini \(2021\)](#). Their key distinction is between *situations of dependency* and *mechanisms of dependency*: between the historically concrete manifestations of peripheral capitalism and the causal processes that reproduce external conditioning. The distinction is decisive for the present chapter. What the classical tradition identified with great force but could not fully formalize was precisely the transmission between external constraint and internal dynamics. The renewed programme begins to close that gap, not by abandoning historical contingency, but by specifying the mechanisms through which external hierarchies shape domestic accumulation, class structure, and political conflict. Around this reformulation, dependency theory has again become a toolkit for the comparative-historical analysis of peripheral capitalisms, rent, and externally conditioned development ([Burchardt et al., 2021](#); [Bizberg, 2021](#)).

2.28 The revival extends beyond epistemological rehabilitation. It has taken substantive form in one of dependency theory’s original sites of elaboration: Santiago, Chile. That matters both symbolically and analytically. What makes the contemporary Santiago-based scholarship important is not simply that it returns to dependency in the abstract, but that it develops new empirical evidence and new diagnoses from within the tradition’s own problematic. [Ahumada and Torres \(2024\)](#) reconstruct dependency theory as a still-productive framework for understanding asymmetries between central and peripheral capitalisms, while [Torres and Ahumada \(2022\)](#) restate the centre–periphery vision as a general lens for interpreting contemporary global capitalism and its rentier tendencies.

2.29 The same agenda then moves into concrete institutional and macro-structural analysis.

Ruiz Bruzzone (2021) analyzes the Chilean pension system as an institutional mechanism of dependent financialization. Ahumada et al. (2025) identify a structural disconnection between surplus capture and productive investment in contemporary Chile, showing how rent extraction can persist without productive upgrading. From the sphere of external insertion, Sossdorf Gonzalez (2024) shows that Chile’s liberal export strategy has generated a stagnant cycle anchored in static comparative advantages, weak technological dynamism, and limited diversification. Taken together, these works operationalize dependency theory to generate historically grounded explanations of financial subordination, rentier accumulation, and externally constrained development in present-day Chile. Dependency theory is being revived in its place of birth as a dense research programme: historically self-conscious, theoretically explicit, and empirically generative.

2.30 The revival also carries a distinct normative horizon, defining development as emancipation from structural subordination. In Ahumada (2023), this appears as a conception centered on non-dependence: for peripheral formations, genuine development requires building the material basis for collective self-determination beyond mere increases in aggregate productivity or income. Arboleda (2023) sharpens this perspective by recovering the memory of the Unidad Popular as the historical conjuncture in which dependency theory, structural planning, and national liberation converged in Chile. Within that horizon, development signified gaining sovereign control over productive assets from oligarchic and transnational capital, redirecting the surplus toward popular needs, worker self-management, and plural property forms.

2.31 That normative content belongs to the framework itself rather than arriving as an external moral add-on. It is part of what made dependency theory politically consequential in Chile, and part of what still gives force to its revival today. Recent work such as Ahumada and Chang (2025) extends rather than abandons this inheritance: the call to recover policy space and rethink the international order continues a tradition whose emancipatory horizon was condensed, with particular clarity, in the experience and defeat of the Unidad Popular.

2.32 What follows takes up that revived horizon at the level of explanation, showing how external constraint, distributive conflict, and macroeconomic dynamics interact within a peripheral economy.

2.6 Epistemological Critique: The Illusion of Causality, Narrative License, and Collider Traps

2.33 The persistent divergence among these explanatory traditions cannot be resolved through historiographical narrative alone. As Blanco and Matute (2018) establish, human causal reasoning is systematically vulnerable to the *illusion of causality*: a robust cognitive bias wherein observers detect strong causal links between co-occurring events even when the statistical contingency is zero or driven by unobserved common causes. In the economic historiography of the Unidad Popular, orthodox monetarist and Austrian accounts observe a conspicuous historical coincidence—the election of a socialist government in late 1970, expansionary fiscal-wage policy in 1971, and the explosion of inflation in 1972–1973. Relying on intuitive heuristics, they assign exclusive causal responsibility to domestic monetary issuance ($g_{M,t} \rightarrow \pi_t$), treating international monetary hierarchy, terms-of-trade shifts, and geopolitical destabilization as irrelevant background noise (Edwards, 2024; Espinosa and Véjar, 2025).

2.34 This causal illusion is reinforced by what Isch et al. (2026) identify as *narrative license* and *overreaching causal claims* in historical social science. Orthodox accounts dress up doctrinal priors as ironclad empirical laws without testing alternative causal DAGs or specifying counterfactual identification paths. For example, Espinosa and Véjar (2025) assert an Austrian Business Cycle mechanism wherein artificial credit expansion generated an unsustainable cap-

ital malinvestment boom. Yet they advance this claim without presenting positive evidence of gross fixed capital deepening (ΔK), ignoring that Chilean fixed machinery investment contracted during 1971 while capacity utilization reached historical peaks. Narrative license allows the theorist to impute unobserved microeconomic mechanisms (capital lengthening) to explain macroeconomic outcomes while dismissing external interventions as quantitatively negligible.

2.35 Methodologically, orthodox accounts fall into an *M-bias collider trap* (Pearl, 2009). In historical narratives of the UP crisis, intermediate symptoms such as retail shortages, food queues, and black markets (C) are conditioned upon as proof of “repressed inflation” caused by price controls. In structural causal terms, retail shortages (C) are an endogenous collider variable caused jointly by domestic demand stimulus (U_1) and private wholesale distribution lockouts, trucking strikes, and hoarding (U_2). Conditioning on empty retail shelves induces a spurious association between public policy (U_1) and private economic sabotage (U_2):

$$U_1 \longrightarrow [C] \longleftarrow U_2 \implies U_1 \not\perp U_2 \mid C \quad (1)$$

By treating retail shortages as a neutral indicator of excess liquidity, orthodox accounts commit a collider stratification error, falsely attributing the supply paralysis orchestrated by private distributor federations exclusively to public deficit monetization.

2.36 To transcend these causal illusions, narrative biases, and collider traps, I adopt the neo-Lakatosian methodology of King and Sznajder (2006). I treat Monetarism (H_1), Austrian Business Cycle Theory (H_2), and Classical Structuralism / Dependency (H_3) as rival scientific research traditions with distinct hard cores, auxiliary assumptions, and falsifiable zero restrictions. I submit the core causal claims of these competing traditions to a formal **Hypothesis Tournament** evaluated across multi-frequency time series data.

3 World Money, Financial Subordination, and the Peripheral Central Bank Balance Sheet

3.1 World Money, the Universal Equivalent, and the Monetary Expression of Value

3.1 In Marxian political economy, money originates as the commodity form of the universal equivalent: the unique social medium whose function is to express the value of all commodities and serve as the general means of payment (Basu, 2022, §3.1). Gold historically occupied this role not through inherent natural properties, but through a social process of market validation, acting as the bearer of abstract human labor.

3.2 The historical transit to credit money transforms this institutional form without dissolving its underlying function. As banking systems developed to intermediate commercial credit, central bank liabilities emerged as the legal anchor of domestic circulation (Basu, 2022, §3.2.3). The commodity anchor disappeared from domestic circulation, but the imperative of social validation remained. Central bank liabilities and bank deposits perform the role that bullion once fulfilled: providing the monetary expression through which private, decentralized labor is validated as social labor. Following the Monetary Expression of Value (MEV) framework (Foley, 1982; Basu, 2022), the value of money links aggregate labor-time to the circulating monetary unit within a national formation.

3.3 Once circulation crosses national borders, however, credit money confronts the world market. While Marx presented world money in the physical form of bullion shedding national garbs (Ivanova, 2013, p. 51), a global credit-money system reproduces the universal equivalent through an international currency hierarchy (Vasudevan, 2009; Ivanova, 2020). National

currencies are stratified by their degree of international acceptability, liquidity, and convertibility. World money is the historically specific form wherein the liabilities of the dominant hegemonic power operate as the universal means of international payment, reserve asset, and unit of account.

3.2 The Bretton Woods Architecture and Peripheral Balance-of-Payments Constraints

3.4 The post-war Bretton Woods architecture codified this currency hierarchy institutionally. By pegging the U.S. dollar to gold at \$35 per ounce while anchoring other currencies to the dollar, the international monetary order formalized the dollar as the operative form of world money (Vasudevan, 2009, p. 483). This arrangement granted the hegemonic center an “exorbitant privilege”: the ability to borrow internationally in its own currency, finance external deficits without reserve constraints, and act as banker to the world (Vasudevan, 2019, pp. 243–244).

3.5 For peripheral economies, this institutional architecture generated an acute structural asymmetry. Peripheral states cannot settle international accounts in their own domestic currencies; external liabilities and essential imports must be discharged in world money. The peripheral circuit of capital confronts an absolute balance-of-payments constraint (Vasudevan, 2016; Abeles, 2013). Domestic credit expansion is bounded by the capacity to command foreign exchange through export earnings, terms of trade, and access to international credit lines. Reserve accumulation constitutes an externally enforced structural prerequisite of economic reproduction (Vasudevan, 2019, p. 244).

3.6 The unilateral closure of the gold window by the United States on August 15, 1971—the Nixon Shock—shattered the fixed-parity anchor of Bretton Woods. While this rupture removed gold convertibility, it preserved dollar centrality, ushering in an era of global floating fiat money and commodity stagflation (Vasudevan, 2009, p. 486). For the nascent socialist project of the Unidad Popular, this systemic shock dismantled the international monetary stability upon which its initial macroeconomic projections rested.

3.3 The Peripheral Central Bank Balance Sheet: From Reserve Adequacy to Leverage Dynamics

3.7 Financial subordination materializes concretely on the balance sheet of the peripheral central bank. Mainstream open-economy models interpret this balance sheet through the lens of “reserve adequacy” and precautionary self-insurance, treating international reserves as an inventory buffer optimized against friction costs to insulate the economy from sudden stops (Jeanne, 2007; Céspedes and Chang, 2024). This normative framing assumes that reserve accumulation represents the voluntary, unconstrained choice of a sovereign optimizing agent.

3.8 In peripheral capitalism, reserve trajectories are the structural outcome of the balance-of-payments constraint, unequal exchange, and imperial financial diplomacy (Vasudevan, 2009; Abeles, 2013). The two sides of the peripheral central bank’s balance sheet reflect fundamentally distinct monetary forms. On the liability side, domestic money represents unanchored credit obligations ($M_{1,t}$ or monetary base H_t) that the financial system owes the public, underwritten by state authority (Basu, 2022, §3.2.3). On the asset side, international reserves (IR_t , measured in USD millions) consist of claims on world money that the peripheral state cannot emit.

3.9 This balance-sheet structure constitutes an active leverage relation:

$$\text{LevR}_t \equiv \frac{M_{1,t}}{e_t \cdot IR_t} = \text{SolvR}_t^{-1} \quad (2)$$

where e_t is the nominal exchange rate (Escudos per USD) and SolvR_t is the Central Bank Solvency Ratio. Financial subordination acquires its operative expression through this leverage relation: movements in SolvR_t govern the structural corridor within which domestic macroeconomic policy can maneuver. The complete formal accounting identities and analytical laws of motion governing this balance-sheet relation are documented in Appendix A.

3.4 Money Endogeneity vs. Exogeneity and the Peripheral Paradox

3.10 Grounding the peripheral balance sheet requires resolving the foundational debate over the causality between bank credit, deposits, and central bank reserves (Rochon and Rossi, 2013; Lavoie, 2014). Mainstream monetarism and orthodox macroeconomics operate under Fractional Reserve Theory (FRT) and loanable-funds axioms: the central bank exogenously fixes the monetary base (H_t), which mechanically dictates bank lending and broad money through a stable fractional-reserve multiplier ($H \rightarrow \text{Deposits} \rightarrow \text{Loans} \rightarrow P$), asserting a unitary elasticity in log levels ($\ln M1_t = \alpha + \beta \ln H_t + \varepsilon_t$, with $\beta = 1, \alpha = 0$). In sharp contrast, Post-Keynesian Endogenous Money Theory (EMT) and Marxian monetary circuits establish that real-world banking operates in reverse causal sequence (Kaldor, 1982; Moore, 1988; Lavoie, 2014):

$$\Delta \text{Bank Loans} \longrightarrow \Delta \text{Deposits} \longrightarrow \Delta \text{Central Bank Reserves}(H) \quad (3)$$

Commercial banks act as price-setters and quantity-takers in credit markets, extending working-capital loans to solvent firms to finance payrolls and production. These loans simultaneously create non-MFI bank deposits out of nothing. The central bank operates as an institutional lender of last resort, compelled to accommodate required reserves ex-post at the policy discount rate to maintain clearance in the interbank payments grid. Under EMT, the traditional money multiplier ($mm_t \equiv M1_t/H_t$) is not an exogenous causal driver but an unstable, ex-post accounting ratio reflecting commercial bank liquidity preferences, cash-drain ratios, and portfolio balance constraints ($\beta \neq 1, \alpha \neq 0$). In peripheral capitalism, credit supply is structuralist and upward-sloping (Palley, 2013; Dow, 2006; Rochon, 1999): commercial banks' willingness to intermediate credit is bounded by external liquidity constraints and foreign exchange exposure.

3.11 This institutional reality establishes what I define as the *Peripheral Paradox*. In core economies issuing world money, the central bank accommodates domestic credit without facing an absolute currency boundary. In the periphery, however, while domestic credit generation is naturally endogenous ($\text{Loans} \rightarrow M1$), the ultimate means of international settlement is bound to an exogenously constrained reserve asset ($e_t \cdot IR_t$). The peripheral central bank must back an endogenously expanding domestic credit pyramid with an externally constrained stock of world money. When external credit embargoes, copper price collapses, or private capital flight drain foreign reserves, the state cannot act as an unconstrained lender of last resort in hard currency; it must choose between contracting domestic activity or suffering balance-sheet insolvency. Under the UP, the nationalized commercial banking system under the State Development Corporation (CORFO) extended defensive working-capital credit to socialized and private enterprises facing distribution bottlenecks and price controls, compelling the Central Bank to monetize enterprise deficits. Domestic money growth thus functioned as an endogenous accommodation valve, driven by structural cost-push pressures rather than discretionary printing press impulses.

3.5 The Peripheral Central Bank Balance Sheet under International Financial Subordination

3.12 To formalize how this balance-sheet contradiction generates macroeconomic crisis, I reformulate the Financial Instability Hypothesis (FIH) of Minsky (1986) for the peripheral central

bank acting as the lender of last resort (LLR). In Minsky’s canonical framework, financial fragility is endogenously generated during prolonged expansions: stability breeds instability as private firms and banks voluntarily increase leverage, transitioning from Hedge to Speculative and Ponzi postures (Kregel, 2008). However, Minsky formulated this taxonomy for private actors in a closed, metropolitan economy possessing complete monetary sovereignty, where an unconstrained central bank (the “Big Bank”) and fiscal authority (the “Big Government”) stabilize the system.

3.13 Transposing this framework to a public central bank in the periphery requires resolving the five theoretical concerns raised by advisors regarding actor logic, profit motives, and exogeneity. First, regarding actor logic and private versus public debt, Foley (2003, p. 21) demonstrates in aggregate cash-flow accounting ($R + D = I + V$) that during acute financial fragility, the state and the central bank as LLR are compelled to absorb the illiquidity and insolvency of private enterprises onto their own balance sheets. The central bank does not lever up out of private profit-seeking or subjective greed; it does so by executing its institutional LLR mandate to prevent payments gridlock and economic breakdown. Second, regarding the objective function, the central bank’s behavior is dictated by the structural balance-sheet identity of reserve solvency:

$$\text{SolvR}_t \equiv \frac{e_t \cdot IR_t}{M_{1,t}} \quad (4)$$

governed by the continuous dynamic law of motion:

$$\Delta \text{SolvR}_t = \frac{CA_t + KA_t}{M_{1,t}} - \text{SolvR}_t \cdot g_{M,t} \quad (5)$$

where CA_t is current account net receipts, KA_t is net capital flows, and $g_{M,t}$ is domestic money growth.

3.14 Third, regarding exogeneity, Chamorro-Futinico (2025) demonstrates that financial instability in peripheral Latin American economies operates as an *externally amplified endogenous fragility*. Minsky’s dynamic cannot be applied to dependent formations without recognizing that external shocks from the center—the collapse of world monetary arrangements, terms-of-trade declines, and credit embargos—serve as the primary triggers detonating domestic balance-sheet fragility. Fourth, regarding motivation, the peripheral central bank enters a *Forced Ponzi* posture not out of speculative euphoria, but under severe institutional duress. When imperial sanctions cut off commercial trade credit, refusing domestic liquidity accommodation would cause immediate real-sector collapse and social starvation. Accommodating the deficits of socialized enterprises is a defensive imperative to maintain societal metabolism.

3.15 Fifth, this reformulation provides the theoretical derivation for the three regimes identified in my empirical Threshold Vector Autoregression (TVAR):

- **Hedge Central Bank** ($\text{SolvR}_t > \hat{\gamma}_{S,1} = 0.21$): Foreign reserve assets (IR_t) comfortably cover domestic liquid liabilities and several months of essential imports. The central bank possesses genuine policy autonomy: expanding domestic credit ($g_{M,t} > 0$) to reactivate idle capacity does not trigger exchange-rate panics because its external backing remains fully credible, yielding zero price pass-through ($\beta_{\text{Hedge}} \approx 0.00\%$).
- **Speculative Central Bank** ($0.14 < \text{SolvR}_t \leq 0.21$): Current export earnings no longer cover external debt service and essential imports simultaneously ($CA_t < 0$). The central bank maintains external settlement by continuously rolling over short-term revolving trade credits ($KA_t > 0$). Reserve erosion strains credibility, generating moderate inflationary pass-through ($\beta_{\text{Spec}} = +0.54\%$).

- **Forced Ponzi Central Bank ($\text{SolvR}_t \leq \hat{\gamma}_{S,2} = 0.14$):** Imperial sanctions unilaterally cancel revolving credit lines ($KA_t \ll 0$), driving liquid reserves below critical import minimums (\$30 million in July 1972). The central bank cannot meet international payment commitments out of current receipts or market borrowing, yet must emit domestic liquidity to prevent the collapse of socialized distribution. This unbacked domestic emission passes directly and discontinuously into hyperinflation ($\beta_{\text{Ponzi}} = +4.54\%$) and currency flight.

3.6 The Four Structural Dimensions of the Unidad Popular Crisis

3.16 Four interconnected structural shocks drove the descent of the Unidad Popular’s Solvency Ratio: First, the August 15, 1971 Nixon Shock dismantled the Bretton Woods system, injecting global commodity inflation directly into Chilean import costs. Second, the secular decline of corporate profitability in the advanced capitalist core (Weisskopf, 1979) motivated imperial retaliation against Chilean copper nationalization. The Nixon administration mobilized an invisible financial blockade that froze Export-Import Bank credits, multilateral loans, and commercial credit lines, precipitating an acute reserve drain.

3.17 Third, this external chokehold opened a relational trade account scissors, defined by the divergence of two distinct economic domains. On the physical and real purchasing power domain, Prebisch-ECLAC Capacity to Import ($M_t^{\text{cap}} \equiv [P_t^{\text{Cu}}/P_{70:11}^{\text{Cu}} \times 100] \times [\text{Exports}_t/\text{Imports}_t^*]$) collapsed from 100.0 to 27.7 in August 1971, measuring the severe contraction in physical import purchasing power resulting from declining copper prices and the embargo on capital goods. This collapse blew open the Real Structural Imbalance ($\Theta_t \equiv M_t/M_t^{\text{cap}} \times 100$, formalizing the Latin American structuralist concept of *desequilibrio estructural externo*) from 105.7 in late 1970 to an unsustainable peak of 283.4 in May 1973, identifying the physical structural imbalance that throttled domestic manufacturing and generated autonomous price pressure. Concurrently, on the nominal and exchange-rate domain, the Relative Price Disparity Index ($\Omega_t \equiv \frac{P_t^{\text{CL}}/P_{70:11}^{\text{CL}}}{(e_t/e_{70:11}) \cdot (P_{W,t}^{\text{US}}/P_{W,70:11}^{\text{US}})} \times 100 \approx 1/\text{RER}_t$) exploded from 100.0 to 190.6 in May 1973 under the official exchange rate peg ($e_t = 0.0122$). Because imperial credit embargoes enforced hard quantity rationing on physical imports in foreign currency, Ω_t diverged from effective import volumes, recording the domestic price disparity maintained by the state’s ethical defense of the social wage.

3.18 Fourth, domestic capitalist resistance compounded this external scissors. The October 1972 Bosses’ Lockout (*Paro Patronal*) and intermediate distribution sabotage paralyzed transport networks and hoarded essential commodities. To prevent industrial paralysis, the central bank was forced to act as lender of last resort to socialized enterprises and state distribution boards, expanding base money to fund basic payrolls. This defensive liquidity expansion crushed the Solvency Ratio below the Ponzi threshold ($\text{SolvR}_t \leq 0.14$) by late 1972, igniting explosive hyperinflation under structural foreign-exchange exhaustion.

4 Data Architecture, Archival Sources, and Variable Construction

4.1 Archival Provenance and Multi-Frequency Research Design

Investigating the macroeconomic trajectory and ultimate breakdown of Chile under the Unidad Popular (UP, 1970–1973) requires an empirical foundation grounded in primary archival ledgers and cross-chapter harmonization. Following the forensic data transparency standards of [Hern-](#)

don et al. (2014), this section documents the provenance of all series, sample exclusions, and variable definitions before presenting historical facts and econometric specifications.

To overcome the temporal aggregation fallacy and closed-system endogeneity bias prevalent in annual narratives (Dornbusch and Edwards, 1990; Edwards, 2024), the research design coordinates two distinct empirical tiers. Low-frequency annual data compress the chronological sequence of political and balance-of-payments shocks, generating the optical illusion that domestic monetary emission was the autonomous prime mover of inflation. A monthly panel is indispensable for isolating causal ordering, balance-sheet transmission, and threshold nonlinearities: in an embattled peripheral open economy facing external credit embargoes, reserve depletion and domestic credit accommodation unfold across weeks, not years.

The high-frequency monthly tier ($N = 250$, 1960:01–1980:12) draws financial and monetary series directly from the Central Bank of Chile’s official database (SieteRestWS). This source provides continuous series for Central Bank base money (H_t), currency in circulation (C_t), the official exchange rate (e_t , Escudos per USD), London Metal Exchange copper prices (P_t^{Cu} , USD/lb), London bullion market gold prices (P_t^{gold} , USD/oz), and the official Consumer Price Index (π_t). For commercial banking transactions, we draw the official un-spliced narrow money series ($M1_t$) and banking multiplier ($m_t \equiv M1_t/H_t$) covering 1966:01–1980:12 ($N = 180$). Gross international reserve assets (IR_t , USD millions) are sourced from the IMF *International Financial Statistics* (IFS, series CHL.TRGV.REVS.USD.M). Real economic activity is measured using disaggregated physical output indices for light industrial manufacturing (Q_t^{manuf}) and enclave mining (Q_t^{mining}) alongside customs export and import volume series constructed by Díaz-Bahamonde (2023) from primary *Estadística Chilena* and Central Bank *Boletín Mensual* ledgers. Following UMass applied econometrics practice, we reject synthetic composite activity indices to prevent collapsing the fundamental real asymmetry of peripheral breakdown: consumer-goods manufacturing expanded by 33.3% in 1971 through idle capacity reactivation, whereas capital-intensive mining grew by only 7.0% under external spare-parts shortages.

The secular annual tier ($T = 91$, 1920–2010) establishes the long-run accumulation and balance-sheet backdrop. Functional income distribution—specifically the wage share (ω_t)—is drawn from the audited inequality ledger of Astorga (2023). Net physical capital stocks across five disaggregated asset categories, the net rate of profit on productive capital, and structural capacity utilization (μ_t) are integrated directly from the harmonized Perpetual Inventory (GPIM) dataset developed in Chapter 2 of this dissertation. Secular balance-of-payments accounts and net international investment positions are drawn from the World Historical Balance of Payments Database (Nievas and Piketty, 2025). Complete archival series codes and metadata are cataloged in Appendix D.

4.2 Variable Construction, Accounting Identities, and Boundary Conditions

To isolate structural mechanisms without imposing ad-hoc statistical filters, variables are operationalized through four core accounting identities:

1. Central Bank Reserve Solvency Ratio (SolvR_t): To capture the foreign-exchange backing of domestic monetary liabilities, we define the Solvency Ratio as:

$$\text{SolvR}_t \equiv \frac{e_t \cdot IR_t}{M1_t \cdot 1000} \quad (6)$$

where e_t is the nominal exchange rate, IR_t is gross international reserves in US dollars, and $M1_t$ is narrow transactions money in thousands of Escudos. For the full 250-month panel (1960–1980), we also construct the base-money variant $\text{SolvR}_t^H \equiv (e_t \cdot IR_t)/(H_t \cdot 1000)$, which

correlates with the narrow-money metric at $\rho = 0.974$. By entering SolvR_{t-1} with a predetermined delay ($d = 1$), the state variable satisfies Hansen (1999) consistency and remains strictly predetermined with respect to innovations at month t .

2. Prebisch Capacity to Import (M_t^{cap}): Following the Prebisch-ECLAC trade framework, the purchasing power of exports in terms of import goods is defined as:

$$M_t^{\text{cap}} \equiv \left(\frac{\text{Exports}_t}{\text{Exports}_{1970:11}} \right) \times \left[\frac{P_t^{\text{Cu}}/P_{1970:11}^{\text{Cu}}}{P_{W,t}^{\text{US}}/P_{W,1970:11}^{\text{US}}} \right] \times 100 \quad (7)$$

where $P_{W,t}^{\text{US}}$ is the U.S. Wholesale Price Index (PPIACO) and Exports_t is real customs export volume from Díaz-Bahamonde (2023). All trade series are benchmarked to November 1970 ($1970 : 11 = 100$), the inaugural month of the UP administration.

3. Real Structural Imbalance Ratio (Θ_t): To measure physical balance-of-payments pressure directly, we define the ratio of effective real customs imports (M_t) to real import purchasing power:

$$\Theta_t \equiv \left(\frac{M_t}{M_t^{\text{cap}}} \right) \times 100 \quad (8)$$

When $\Theta_t > 100\%$, physical import volumes exceed real export purchasing power, generating a structural foreign-exchange deficit that must be financed through foreign credit lines or by depleting Central Bank reserves. To resolve a physical data distortion caused by the nationwide customs strike of April 1973, import values for that month are linearly interpolated between March and May 1973.

4. Integration Order Governance as Specification Gate: Following UMass applied econometrics guidelines, no variable enters dynamic time-series regressions solely on theoretical grounds. We subject all candidate series to four complementary unit root tests (ADF, DF-GLS, Phillips-Perron, and KPSS). All variables reject unit root non-stationarity in first differences ($p < 0.01$), establishing $d_{\text{max}} = 1$ across the candidate system and authorizing their inclusion in the Toda–Yamamoto levels VAR and Threshold VAR architectures. Complete test statistics and critical values are reported in Appendix C.

5 Historical Background, Stylized Facts, and Empirical Results

5.1 Historical Background: The Rise of the Unidad Popular

5.2 Stylized Facts of Peripheral Accumulation, Distribution, and External Constraints

+

5.3 Empirical Exercise: *Lakatosian* Granger Causality Tournament

5.3.1 Econometric Framework: Toda–Yamamoto Levels Augmentation

5.3.2 Univariate Stationarity and Integration Pre-testing Battery ($d_{\max} = 1$)

5.3.3 Sequential Tournament Architecture and Hypothesis Testing Rules

5.3.4 Empirical Findings

5.3.5 Residual Diagnostics and Dynamic Stability

5.4 Empirical Exercise: Non-Linear Dynamics under Peripheral Central Bank Solvency. A Threshold VAR Approach

5.4.1 The Threshold VAR Specification and Structural Identification

5.4.2 Endogenous Threshold Estimation and Residual Bootstrap Linearity Testing

5.4.3 Regime Parameter Estimates and System Stability Forensics

5.4.4 Dynamic Generalized Impulse Response Functions (GIRFs)

5.4.5 Formal Multiplier Difference Hypothesis Tests

6 Political Economy Discussion, Global Implications, and Concluding Synthesis

6.1 The Political Economy of Peripheral Solvency and Subordinated Money

5.1 The econometric findings in Section 5 dismantle the foundational premise shared by orthodox interpretations of the Unidad Popular: the assumption that money growth is an unconstrained, exogenous policy choice. By showing that inflation and currency depreciation systematically preceded base money expansion ($\pi_{t-1} \rightarrow g_{M,t}$), and that the inflationary pass-through of monetary growth is strictly state-dependent on central bank reserve solvency (SolvR_t), my analysis opens the “black box” of peripheral monetary transmission. Central Bank money creation operated as the institutional accommodation valve through which the monetary authority, acting as lender of last resort (LLR), mediated the collision between domestic distributive commitments and an externally enforced foreign-exchange blockade.

5.2 In a peripheral capitalist economy, the central bank operates under what I define as the *Peripheral Paradox* (Section 3.4). While the domestic credit pyramid is naturally endogenous—commercial lending creates deposits, which the central bank must accommodate to prevent liquidity gridlock—the ultimate means of international settlement is exogenously rationed by access to world money ($e_t \cdot IR_t$). When external commercial credit lines are severed and copper receipts are tied up in litigation, the peripheral central bank cannot freely print international purchasing power. As reserves drain, the domestic monetary base (H_t) becomes structurally decoupled from foreign exchange backing. The central bank is placed in an impossible institutional bind: either it refuses liquidity to state-owned enterprises and private firms facing import cost explosions, triggering immediate mass bankruptcies and physical output collapse; or it acts as the lender of last resort, injecting emergency liquidity that, in the absence of foreign reserves to

anchor the currency, accelerates domestic hyperinflation. The descent of the central bank’s own balance sheet into a Forced Ponzi regime ($\text{SolvR}_m \leq 0.14$), transposing the Minsky-Foley framework (Foley, 2003; Chamorro-Futinico, 2025) to peripheral central bank solvency, registered the structural exhaustion of peripheral solvency under external financial subordination.

5.3 This institutional reality exposes the flaw in the orthodox treatment of retail shortages, rationing, and black markets. In the monetarist narrative of Edwards (2024), shortages are treated as unambiguous evidence of “repressed inflation” caused by price controls. In structural causal terms, however, retail shortages (C) constitute an *endogenous collider variable* (Pearl, 2009) caused jointly by wage-led demand stimulus (U_1) and private wholesale distribution sabotage, trucking strikes, and hoarding (U_2):

$$U_1 \longrightarrow [C] \longleftarrow U_2 \implies U_1 \not\perp\!\!\!\perp U_2 \mid C \quad (9)$$

By conditioning on empty store shelves without modeling the October 1972 Bosses’ Lockout (*Paro de Octubre*) or the deliberate withholding of inventories by the wholesale distribution monopoly (SDI), orthodox accounts commit an M-bias collider stratification error. They mistake the physical symptoms of organized class struggle and distribution sabotage for the mechanical consequences of central bank balance sheets, generating a powerful causal illusion that conceals the political economy of destabilization (Blanco and Matute, 2018; Casals, 2021).

6.2 The Unidad Popular in the Global Structural Crisis of the 1970s

5.4 The crisis of the Unidad Popular cannot be understood in national isolation. It was the peripheral expression of a global structural crisis of capitalism that unfolded across the early 1970s. The long postwar boom in the advanced capitalist core was exhausting itself: corporate profit rates in the United States and Western Europe suffered a protracted secular decline (Weiskopf, 1979; Bowles et al., 1983), labor militancy intensified, and the institutional foundations of Atlantic Fordism began to fracture. When President Richard Nixon closed the gold window on August 15, 1971, he unilaterally dissolved the gold-dollar convertibility anchor of the Bretton Woods system. This hegemonic shock exported U.S. stagflation directly to the periphery while reconfiguring the international currency hierarchy.

5.5 For Chile, the timing was catastrophic. The Vuskovic economic program had been designed in late 1970 under the explicit assumption of a stable, fixed-exchange-rate international monetary regime backed by ample inherited reserves. Nine months into the administration, the institutional anchor of world money vanished. U.S. inflation drove up the dollar prices of Chilean imports, while the Nixon administration’s directive to “make the economy scream” mobilized multilateral lending freezes (World Bank, IDB, Eximbank) and private commercial credit boycotts. The hegemonic state, facing its own domestic profitability crisis, deployed external financial subordination as an offensive instrument of imperial discipline against a peripheral nation exercising sovereignty over its natural resources (Harmer, 2011; Arboleda, 2023). The Unidad Popular fell when its redistributive commitments collided directly with the hard boundaries of the international currency hierarchy during an acute global structural crisis.

6.3 Policy Implications for Contemporary Peripheral Sovereignty

5.6 This historical reassessment yields fundamental analytical and policy lessons for contemporary peripheral formations attempting progressive or socialist developmental transitions. The Chilean experience provides a definitive critique of naive extensions of Modern Monetary Theory (MMT) to the global South. In an asymmetric international monetary hierarchy, sovereign

authority over domestic currency issue does not confer command over world money (Vasudevan, 2009; Alami et al., 2023). A peripheral government cannot simply finance developmental programs through central bank monetization without confronting the structural ceiling of foreign-exchange reserve solvency.

5.7 Peripheral monetary autonomy is strictly conditioned by the width of the external corridor. Sustaining developmental transitions requires active macroprudential management of foreign exchange, including mandatory surrender of export revenues, differentiated exchange-rate bands, and permanent, counter-cyclical capital controls to insulate domestic policy from international liquidity cycles (Ffrench-Davis, 2018; Vasudevan, 2019). High nominal commodity export revenues are also insufficient to secure development if the physical composition of imports is vulnerable to external leverage. Peripheral planning must prioritize strategic self-sufficiency in non-substitutable capital equipment, replacement parts, and domestic food security to prevent foreign-exchange blockades from paralyzing the domestic productive apparatus.

6.4 Concluding Synthesis: Toward a Progressive Research Programme

5.8 For more than fifty years, the historiography of the Unidad Popular has operated as an ideological weapon—a cautionary tale constructed by monetarist and Austrian orthodoxy to declare the impossibility of socialist transformation. By subjecting these claims to a formal Neo-Lakatosian Hypothesis Tournament, this chapter demonstrates that the orthodox narrative collapses under econometric scrutiny. First, the Monetarist hypothesis (H_1) fails: prior price shocks unidirectionally forced emergency monetary accommodation on the central bank balance sheet. Second, the Austrian ABCT framework (H_2) fails: the 1971 output surge represented a Keynesian-Structuralist reactivation of idle industrial capacity ($u = 98.4\%$) alongside contracting fixed capital formation ($\Delta K = -2.3\%$), ruling out credit-driven malinvestment. Third, the Structural-Dependency thesis (H_3) holds: the inflationary pass-through of monetization is strictly governed by central bank solvency thresholds (SolvR_t). Domestic liquidity is absorbed without inflation when reserves are solvent ($\text{IRF}_\pi = 0.00\%$), but detonates into hyperinflation once imperial blockades and reserve exhaustion push the central bank balance sheet into the Forced Ponzi regime ($+4.54\%$ at $h = 0$).

5.9 Combining the relational thickness of a canonical case study with regime-switching econometrics, this chapter advances a *progressive problem shift* within the dependency research programme (Kvangraven, 2021; King and Sznajder, 2006). The analytical framework formalized here—the Solvency Ratio threshold model, the multi-frequency tournament architecture, and the decomposition of import composition constraints—provides a generative toolkit for analyzing peripheral financial subordination across historical and contemporary settings. From the structural crises of the Southern Cone in the 1970s and 1980s (Argentina, Peru) to contemporary debt traps in the global South, the central bank balance sheet remains the contested terrain where external financial subordination enters domestic reproduction. Reclaiming sovereignty in the periphery requires understanding that the struggle for economic democracy cannot stop at the borders of the nation-state; it must confront and transform the hierarchical architecture of the world monetary system.

References

- Abeles, M. (2013). Duncan Foley’s circuit of capital model for an open economy. In *Social Fairness and Economics*, pages 228–253. Routledge.
- Afonso, A., Baxa, J., and Slavík, M. (2018). Fiscal developments and financial stress: Does financial stress matter for fiscal policy? *Empirical Economics*, 55(4):1851–1882.
- Ahumada, J. M. (2023). Bringing freedom back to developmentalism: Industrialisation as national independence. *Cambridge Journal of Economics*, 47(6):1037–1056.
- Ahumada, J. M. and Chang, H.-J. (2025). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- Ahumada, J. M., Méndez, L., Sossdorf, F., and Jara, R. (2025). Rentier capitalism in the chilean economy: disconnection between surplus capture and productive investment. *New Political Economy*, pages 1–22.
- Ahumada, J. M. and Torres, M. (2024). Dependency theory: an underdevelopmental thought from latin america to the entire world. In *Global Handbook of Inequality*, pages 1–21. Springer.
- Alami, I., Alves, C., Bonizzi, B., Kaltenbrunner, A., Koddenbrock, K., Kvangraven, I., and Powell, J. (2023). International financial subordination: a critical research agenda. *Review of International Political Economy*, 30(4):1360–1386.
- Aldunate, F., González, F., and Prem, M. (2022). The effects of U.S. banking covert action during Allende’s Chile. *Journal of Financial Economics*. GAP FLAG: confirm journal, volume, pages.
- Arboleda, M. (2023). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- Astorga, P. (2023). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- Balke, N. S. (2000). Credit and economic activity: credit regimes and nonlinear propagation of shocks. *Review of Economics and Statistics*, 82(2):344–349.
- Bambirra, V. (1971). *Tipología de la dependencia en América Latina*. CESO, Universidad de Chile, Santiago.
- Baran, P. A. (1957). *The political economy of growth*. Monthly Review Press, New York.
- Basu, D. (2022). *The Logic of Capital: An Introduction to Marxist Economic Theory*. Cambridge University Press, Cambridge.
- Bautista, M., González, F., Martínez, M., Muñoz, P., and Prem, M. (2023). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- Bitar, S. (2013). *Chile 1970–1973: asumir la historia para construir el futuro*. Peña Lillo / Ediciones América del Sur, Santiago. Originally published 1979.
- Bizberg, I. (2021). Post commodity boom crises in Latin America. In Madariaga, A. and Palestini, S., editors, *Dependent Capitalisms in Contemporary Latin America and Europe*, pages 131–155. Palgrave Macmillan.
- Blanco, F. and Matute, H. (2018). The illusion of causality: a cognitive bias underlying pseudoscience. In Kaufman, A. B. and Kaufman, J. C., editors, *Pseudoscience: The Conspiracy Against Science*, pages 45–76. MIT Press, Cambridge, MA.
- Bockman, J. (2019). Democratic socialism in chile and peru: Revisiting the “chicago boys” as the origin of neoliberalism. *Comparative Studies in Society and History*, 61(3):654–679.
- Bowles, S., Gordon, D. M., and Weisskopf, T. E. (1983). *Beyond the Waste Land: A Democratic Alternative to Economic Decline*. Anchor Press/Doubleday, Garden City, NY.

- Braun, J., Braun, M., Briones, I., Díaz, J., Lüders, R., and Wagner, G. (2000). *Economía chilena 1810–1995: Estadísticas históricas*. Documento de Trabajo 187, Instituto de Economía, Pontificia Universidad Católica de Chile.
- Burchardt, H.-J., Dietz, K., and Warnecke-Berger, H. (2021). Dependency, rent, and the failure of neo-extractivism. In Madariaga, A. and Palestini, S., editors, *Dependent Capitalisms in Contemporary Latin America and Europe*, pages 207–229. Palgrave Macmillan.
- Caputo, O. and Pizarro, R. (1970). *Imperialismo, dependencia y relaciones económicas internacionales*. CESO, Universidad de Chile, Santiago.
- Cárdenas, E., Lana, L., and Seabra, R. (2022). *El CESO y la dependencia: orígenes, debates y legado*. CLACSO. GAP FLAG: confirm publisher and editors.
- Cardoso, F. H. and Faletto, E. (1979). *Dependency and development in Latin America*. University of California Press, Berkeley. Originally published in Spanish, 1969.
- Casals, M. (2021). The chilean counter-revolution: Roots, dynamics and legacies of mass mobilisation against the unidad popular. *Radical Americas*, 6(1):1–17.
- Céspedes, L. F. and Chang, R. (2024). Optimal foreign reserves and central bank policy under financial stress. *American Economic Journal: Macroeconomics*, 16(3):230–267.
- Chamorro-Futinicó, J. C. (2025). Boom and bust: Dynamics of systemic financial instability in financialized emerging economies. *Review of Political Economy*, Ahead-of-print:1–25.
- Charnysh, V., Finkel, E., and Gehlbach, S. (2023). Historical political economy: core concepts and new directions. In Jenkins, J. A. and Rubin, J., editors, *The Oxford Handbook of Historical Political Economy*. Oxford University Press. GAP FLAG: confirm if this is a standalone chapter or part of handbook introduction; verify year 2023 vs 2024.
- Clark, T. D. (2017). Rethinking Chile’s “Chicago Boys”: neoliberal technocrats or revolutionary vanguard? *Third World Quarterly*, 38(6):1350–1365.
- Cuesta, J. I., Gallego, F., and González, F. (2015). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- De Castro, S. (1992). *El Ladrillo: bases de la política económica del gobierno militar chileno*. Centro de Estudios Públicos, Santiago. Originally written 1973; published 1992.
- De Vylder, S. (1976). *Allende’s Chile: the political economy of the rise and fall of the Unidad Popular*. Cambridge University Press, Cambridge.
- Díaz, J., Lüders, R., and Wagner, G. (2016). *Chile 1810–2010: La República en Cifras*. Ediciones Universidad Católica de Chile, Santiago, Chile.
- Díaz-Bahamonde, J. (2023). An index of monthly economic activity for chile, 1927–2021: A proposal. *Documento de Trabajo, Instituto de Economía UC*.
- Dornbusch, R. and Edwards, S. (1990). *The Macroeconomics of Populism in Latin America*. University of Chicago Press.
- Dornbusch, R. and Edwards, S. (1991). The macroeconomics of populism. In Dornbusch, R. and Edwards, S., editors, *The Macroeconomics of Populism in Latin America*, pages 7–13. University of Chicago Press, Chicago.
- Dos Santos, T. (1970). The structure of dependence. *American Economic Review*, 60(2):231–236.
- Dow, S. C. (2006). Endogenous money: Structuralist. In Arestis, P. and Sawyer, M., editors, *A Handbook of Alternative Monetary Economics*, pages 35–51. Edward Elgar, Cheltenham, UK.
- Edwards, S. (2024). Inflation and the corruption of currency in Latin America: Chile, 1970–1973. *Journal of Policy History*, 36(4):386–399.
- Espinosa, A. (2021). *The economic impossibility of socialism: an Austrian analysis of Allende’s Chile*. Routledge. GAP FLAG: confirm publisher.

- Espinosa, J. G. and Zimbalist, A. S. (1978). *Economic democracy: workers' participation in Chilean industry, 1970–1973*. Academic Press, New York.
- Espinosa, V. I. and Véjar, V. (2025). Macroeconomic populism in Chile: Allende and the recession of 1973. *Revista de Historia Económica - Journal of Iberian and Latin American Economic History*, 43(2):309–332.
- Ffrench-Davis, R. (2018). *Reformas económicas en Chile 1973–2017*. Taurus, Santiago.
- Fischer, K. (2009). The influence of neoliberals in chile before, during, and after pinochet. *The road from Mont Pèlerin: The making of the neoliberal thought collective*, pages 305–346.
- Foley, D. K. (1982). The value of money, the value of labour-power and the Marxian transformation problem. *Review of Radical Political Economics*, 14(2):37–47.
- Foley, D. K. (2003). Financial fragility in developing economies. *Working Paper Series, Department of Economics, New School University*.
- Frank, A. G. (1967). *Capitalism and underdevelopment in Latin America: historical studies of Chile and Brazil*. Monthly Review Press, New York.
- Girardi, D. and Bowles, S. (2018). Institution shocks and economic outcomes: Allende's election, Pinochet's coup and the Santiago stock market. *Journal of Development Economics*, 134:16–27.
- Gonçalves, S. and Kilian, L. (2004). Bootstrapping autoregressions with conditional heteroskedasticity of unknown form. *Journal of Econometrics*, 123(1):89–120.
- González, F. (2013). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- González, F. and Prem, M. (2026). [title to be confirmed]. GAP FLAG: title, venue, and year unconfirmed — researcher must verify before submission.
- González, F., Prem, M., and Urzúa, F. (2020). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- González, F. and Vial, J. (2021). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- Griffith-Jones, S. (1981). *The role of finance in the transition to socialism*. Frances Pinter, London.
- Hansen, B. E. (1999). Threshold effects in non-dynamic panels: Estimation, testing, and inference. *Journal of Econometrics*, 93(2):345–368.
- Harmer, T. (2011). *Allende's Chile and the Inter-American Cold War*. University of North Carolina Press, Chapel Hill.
- Herndon, T., Ash, M., and Pollin, R. (2014). Does high public debt consistently stifle economic growth? a critique of reinhart and rogoft. *Cambridge Journal of Economics*, 38(2):257–279.
- Isch, C., Dörr, T., Fasching, N., Jennings, G., and Watts, D. J. (2026). Quantifying the prevalence and impact of overreaching causal claims in social science. *Nature Human Behaviour*. Advance online publication.
- Ivanova, M. N. (2013). The dollar as world money. *Science & Society*, 77(1):44–71.
- Ivanova, M. N. (2020). Marx's theory of money: A reappraisal in the light of unconventional monetary policy. *Review of Radical Political Economics*, 52(1):137–151.
- Jaimovich, A. and Toledo-Concha, E. (2021). Failed agrarian reform and the origins of contemporary conflict: evidence from Chile. *The Economic Journal*.
- Jeanne, O. (2007). International reserves in emerging market countries: too much of a good thing? *Brookings papers on Economic activity*, 2007(1):1–79.
- Jenkins, J. A. and Rubin, J., editors (2024). *The Oxford Handbook of Historical Political Economy*. Oxford University Press.

- Kaldor, N. (1982). *The Scourge of Monetarism*. Oxford University Press, Oxford.
- Kay, C. (1978). Agrarian reform and the class struggle in Chile. In *Chile after 1973*. Latin America Bureau, London. GAP FLAG: confirm booktitle and publisher.
- Kay, C. (2010). *Latin American theories of development and underdevelopment*. Routledge, London.
- King, L. P. and Sznajder, A. (2006). The state-led transition to liberal capitalism: Neoliberal, organizational, world-systems, and social structural explanations of Poland's economic success. *American Journal of Sociology*, 112(3):751–801.
- Koop, G., Pesaran, M. H., and Potter, S. M. (1996). Impulse response analysis in nonlinear multivariate models. *Journal of Econometrics*, 74(1):119–147.
- Kregel, J. (2008). Minsky's cushions of safety, systemic risk and the crisis in the subprime mortgage market. Public Policy Brief 93, Levy Economics Institute of Bard College.
- Kvangraven, I. H. (2021). Beyond the stereotype: Restating the relevance of the dependency research programme. *Development and change*, 52(1):76–112.
- Lakatos, I. (1978). *The Methodology of Scientific Research Programmes: Philosophical Papers Volume 1*. Cambridge University Press, Cambridge.
- Larraín, F. and Meller, P. (1991). The socialist-populist Chilean experience, 1970–1973. In Dornbusch, R. and Edwards, S., editors, *The Macroeconomics of Populism in Latin America*, pages 175–221. University of Chicago Press, Chicago.
- Lavoie, M. (2014). *Post-Keynesian Economics: New Foundations*. Edward Elgar Publishing, Cheltenham, UK.
- Leiva, F. I. (2008). *Latin American neostructuralism: the contradictions of post-neoliberal development*. University of Minnesota Press, Minneapolis.
- Loveman, B. (1971). *Antecedentes para el estudio del movimiento campesino chileno: pliegos de peticiones, huelgas y sindicatos agrícolas, 1932–1966*. Instituto de Capacitación e Investigación en Reforma Agraria (ICIRA), Santiago. 2 vols.
- Loveman, B. (1976). *Struggle in the Countryside: Politics and Rural Labor in Chile, 1919–1973*. Indiana University Press, Bloomington, IN.
- Madariaga, A. and Palestini, S., editors (2021). *Dependent capitalisms in contemporary Latin America and Europe*. Palgrave Macmillan.
- Maillet, A., Toro, S., Olivares, A., and Rodríguez, M. I. (2016). Los monjes fuera del monasterio: CIEPLAN y su producción intelectual durante cuatro décadas de política chilena. *Política. Revista de Ciencia Política*, 54(1):189–218.
- Marini, R. M. (1973). *Dialéctica de la dependencia*. Era, México.
- Meller, P. (2000). *The Unidad Popular and the Pinochet dictatorship: a political analysis*. Macmillan, Basingstoke. GAP FLAG: confirm exact title — may be chapter in edited volume or different title.
- Minsky, H. P. (1986). *Stabilizing an Unstable Economy*. Yale University Press, New Haven.
- Moore, B. J. (1988). *Horizontalists and Verticalists: The Macroeconomics of Credit Money*. Cambridge University Press, Cambridge.
- Morris, J. O. and Oyaneder, R. (1962). *Afiliación y finanzas sindicales en Chile, 1932–1959*. INSORA, Universidad de Chile, Santiago.
- Murphy, E. (2015). *For a Proper Home: Housing Rights in the Margins of Urban Chile, 1960–2010*, volume 349. University of Pittsburgh Press, Pittsburgh, PA.
- Nievas, G. and Piketty, T. (2025). The world balance of payments database: 1900–2024. *World Inequality Lab Working Paper*, 2025/04.
- Osorio, S. and Polanco, D. (2021). Tasas de afiliación sindical y tamaño promedio de los

- sindicatos en chile, 1932–2010. Minuta 2, Repositorio de Estadísticas Sindicales (RES).
- Palley, T. I. (2013). Horizontalists, verticalists, and structuralists: The theory of endogenous money. In *Financialization: The Macroeconomics of Finance Capital*, pages 109–126. Palgrave Macmillan, London.
- Palma, J. G. (2016). The “dependency school” and its aftermath: why latin america’s critical thinking switched from one type of absolute certainties to another’. *Handbook of alternative theories of economic development*, pages 386–415.
- Pearl, J. (2009). *Causality: Models, Reasoning, and Inference*. Cambridge University Press, Cambridge, 2nd edition.
- Petras, J. F. and Morley, M. H. (1975). *The United States and Chile: imperialism and the overthrow of the Allende government*. Monthly Review Press, New York.
- Ramos, J. (1980). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- Ramos, S. (1979). [title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- Rochon, L.-P. (1999). *Credit, Money, and Production: An Alternative Post-Keynesian Approach*. Edward Elgar Publishing, Cheltenham, UK.
- Rochon, L.-P. and Rossi, S. (2013). Endogenous money: the evolutionary versus revolutionary views. *Review of Keynesian Economics*, 1(2):210–229.
- Rodríguez-Asensio, D. and López-García, A. M. (2026). Transmission of macroeconomic shocks in housing markets: A threshold VAR approach. *Journal of Real Estate Finance and Economics*.
- Rodríguez Weber, J. E. (2018). Alta desigualdad en América Latina: desde cuándo y por qué. *Documentos de Trabajo en Historia Económica*.
- Roncaglia, A. (2006). Why should economists study the history of economic thought? *The European Journal of the History of Economic Thought*, 13(1):1–18.
- Ruiz Bruzzone, C. (2021). Financialization and peripheral business power in the Chilean pension system. In Madariaga, A. and Palestini, S., editors, *Dependent Capitalisms in Contemporary Latin America and Europe*, pages 231–257. Palgrave Macmillan.
- Runge, J., Nowack, P., Kretschmer, M., Flaxman, S., and Sejdinovic, D. (2019). Detecting and quantifying causal associations in large nonlinear time series datasets. *Science Advances*, 5(11):eaau4996.
- Sideri, S., editor (1979). *Chile 1970–73: economic development and its international setting: self-criticism of the Unidad Popular government’s policies*. Martinus Nijhoff, The Hague.
- Silva, P. (1991). Technocrats and politics in Chile: from the Chicago Boys to the CIEPLAN Monks. *Journal of Latin American Studies*, 23(2):385–410.
- Sims, C. A. and Zha, T. (1999). Error bands for impulse responses. *Econometrica*, 67(5):1113–1155.
- Sklair, L. (2000). The transnational capitalist class and the discourse of globalisation. *Cambridge Review of International Affairs*, 14(1):67–85.
- Sossdorf Gonzalez, F. (2024). [trade paper title to verify]. Compatibility placeholder for consolidated dissertation build; verify bibliographic details before submission.
- Stallings, B. (1978). *Class conflict and economic development in Chile, 1958–1973*. Stanford University Press, Stanford.
- Stallings, B. and Zimbalist, A. (1975). The political economy of Chilean underdevelopment. *Latin American Perspectives*, 2(4):69–88. GAP FLAG: confirm volume, number, pages.
- Sweezy, P. M. and Magdoff, H. (1974). *Revolution and counter-revolution in Chile*. Monthly

- Review Press, New York.
- Torres, M. and Ahumada, J. M. (2022). Las relaciones centro-periferia en el siglo xxi. *El trimestre económico*, 89(353):151–195.
- Valdés, J. G. (1995). *Pinochet’s economists: the Chicago school of economics in Chile*. Cambridge University Press, Cambridge.
- Vasudevan, R. (2009). From the gold standard to the floating dollar standard: An appraisal in the light of Marx’s theory of money. *Review of Radical Political Economics*, 41(4):473–491.
- Vasudevan, R. (2016). Financialization, distribution and accumulation: a circuit of capital model with a managerial class. *Metroeconomica*, 67(2):397–428.
- Vasudevan, R. (2019). Money, power, and capitalism: Marx’s theory of money and the contemporary state-credit standard. In Gupta, S., Basu, M. M., and Babones, S., editors, *Karl Marx’s Life, Ideas, and Influences: A Critical Examination on the Bicentenary*, pages 229–246. Palgrave Macmillan, Cham.
- Weisskopf, T. E. (1979). Marxian crisis theory and the rate of profit in the postwar u.s. economy. *Cambridge Journal of Economics*, 3(4):341–378.
- Winn, P. (1986). *Weavers of revolution: the Yarur workers and Chile’s road to socialism*. Oxford University Press, New York.
- Zavaleta Mercado, R. (1974). El poder dual en América Latina: estudio de los casos de Bolivia y Chile. *Cuadernos Políticos*.

Appendix A Formalization of the Central Bank Solvency Ratio and Balance-of-Payments Accounting

A.1 This appendix develops the formal balance-sheet accounting, the exact discrete law of motion, and the structural constraint-binding conditions governing the central bank **Solvency Ratio** (SolvR_t). Every accounting step is an exact discrete identity; no linear approximation or continuous-time distortion enters the derivation.

A.1 The Solvency Ratio in Nominal Terms

A.2 All variables are denominated in current domestic currency (Chilean pesos/escudos, CLP) at time t , with the nominal exchange rate convention defined as e_t (CLP per USD), such that a nominal depreciation corresponds to $e_t \uparrow$.

A.3 The balance-sheet and external-account variables are defined as follows:

- M_t : domestic monetary obligations (CLP), representing the liquid liabilities of the consolidated domestic financial system underwritten by the central bank.
- IR_t : stock of gross international reserves (USD), representing claims denominated in world money.
- e_t : nominal exchange rate (CLP per USD).
- $F_t \equiv e_t \cdot IR_t$: stock of world money held by the central bank, valued in domestic currency (CLP).
- $\text{BoP}_t \equiv \Delta F_t$: net balance-of-payments flow (CLP).
- $g_{M,t} \equiv \Delta M_t / M_t$: simple growth rate of domestic money obligations.
- $g_{F,t} \equiv \Delta F_t / F_t = \text{BoP}_t / (e_t \cdot IR_t)$: simple growth rate of the world-money asset base.
- X_t : export volume of primary and manufactured goods.
- Q_t : import volume of intermediate, consumption, and capital goods.
- $P_{Q,t}^{USD}$: import price index in foreign currency (USD).
- $\text{ToT}_t \equiv P_{X,t}^{USD} / P_{Q,t}^{USD}$: terms of trade.
- D_t^{ext} : stock of external debt (USD).
- r_t : average nominal interest rate on external debt.
- FDI_t : net foreign direct investment inflows (USD).
- Π_t : net foreign profit and dividend remittances (USD).
- T_t : net current external transfers (CLP).

Import volumes carry the symbol Q_t throughout to prevent notation collisions with domestic money M_t . Discrete growth rates are evaluated on the contemporaneous base to maintain algebraic exactness.

A.4 I define the central bank Solvency Ratio (SolvR_t) in nominal terms as:

$$\text{SolvR}_t \equiv \frac{e_t \cdot IR_t}{M_t} = \frac{F_t}{M_t} \quad (10)$$

The Solvency Ratio measures the stock of world-money assets held by the central bank per unit of unanchored domestic credit obligations underwritten by the monetary authority. Its algebraic inverse, $\text{LevR}_t \equiv 1/\text{SolvR}_t = M_t/(e_t \cdot IR_t)$, expresses the corresponding balance-sheet leverage ratio. I center the analytical framework on the Solvency Ratio because it directly formalizes central bank reserve backing and operationalizes the transposition of Minsky's financial postures to the peripheral central bank's own balance sheet (Foley, 2003; Chamorro-Futinico, 2025) (Section 3.5).

A.5 Taking continuous growth rates of Eq. (10) yields the continuous growth decomposition:

$$g_{\text{SolvR},t} = g_{F,t} - g_{M,t} = g_{e,t} + g_{IR,t} - g_{M,t} \quad (11)$$

Equation (11) isolates three distinct structural channels driving peripheral balance-sheet solvency:

1. **External Balance-of-Payments Channel** ($g_{IR,t}$, expanding solvency): Export revenues and net foreign capital inflows accumulate foreign exchange reserves, bolstering central bank solvency. Current account deficits, foreign debt service, and transnational profit repatriation drain reserves, pulling SolvR_t downward.
2. **Valuation Channel** ($g_{e,t}$, expanding nominal solvency): Nominal currency depreciation ($e_t \uparrow$) increases the domestic-currency accounting valuation of foreign assets ($F_t = e_t \cdot IR_t$). While this mechanically cushions SolvR_t in domestic currency units, it carries acute real macroeconomic costs by inflating imported intermediate input costs and triggering domestic wage-price defense spirals.
3. **Domestic Dilution Channel** ($g_{M,t}$, compressing solvency): Broad money creation, fiscal monetization, and emergency Lender-of-Last-Resort liquidity expand domestic obligations from below, continuously diluting the world-money backing of the monetary base.

A.2 Current and Capital Account Decomposition

A.6 The evolution of the central bank's foreign-currency asset base is governed by the external balance of payments. In domestic currency terms, the current account (CA_t) and capital account (KA_t) are specified as:

$$CA_t = e_t \cdot P_{Q,t}^{USD}(\text{ToT}_t \cdot X_t - Q_t) - e_t \cdot r_t \cdot D_t^{\text{ext}} + T_t \quad (12)$$

$$KA_t = e_t \cdot \Delta D_t^{\text{ext}} + e_t \cdot FDI_t - e_t \cdot \Pi_t \quad (13)$$

The sum of the current and capital accounts determines the net balance-of-payments flow (BoP_t), which dictates the net change in world-money holdings:

$$\Delta F_t \equiv \Delta(e_t \cdot IR_t) = CA_t + KA_t \equiv \text{BoP}_t \quad (14)$$

A.3 Derivation of the Discrete Solvency Law of Motion

A.7 From Eq. (10), the total stock of world-money assets held by the central bank decomposes into the product of the Solvency Ratio and domestic monetary obligations:

$$F_t = \text{SolvR}_t \cdot M_t \quad (15)$$

Applying the exact discrete product rule $\Delta(A_t B_t) = B_t \Delta A_t + A_{t-1} \Delta B_t$ with $A_t = \text{SolvR}_t$ and $B_t = M_t$ yields:

$$\Delta F_t = M_t \cdot \Delta \text{SolvR}_t + \text{SolvR}_{t-1} \cdot \Delta M_t \quad (16)$$

Dividing Eq. (16) through by M_t and substituting $\Delta F_t = \text{BoP}_t$ from Eq. (14):

$$\frac{\text{BoP}_t}{M_t} = \Delta \text{SolvR}_t + \text{SolvR}_{t-1} \cdot \frac{\Delta M_t}{M_t} \quad (17)$$

Rearranging terms isolates the exact discrete law of motion for the Solvency Ratio:

$$\boxed{\Delta \text{SolvR}_t = \frac{\text{BoP}_t}{M_t} - \text{SolvR}_{t-1} \cdot g_{M,t}} \quad (18)$$

where $g_{M,t} \equiv \Delta M_t / M_t$ is the simple growth rate of domestic monetary liabilities.

A.8 Equation (18) delivers an intuitive and powerful economic decomposition:

- The first term, BoP_t / M_t , represents the **external liquidity injection rate**—net foreign exchange inflows scaled directly by the existing domestic credit pyramid. Balance-of-payments surpluses expand the asset base and bolster solvency.
- The second term, $\text{SolvR}_{t-1} \cdot g_{M,t}$, represents the **domestic dilution drag**. Rapid domestic money growth ($g_{M,t} > 0$) dilutes the inherited backing ratio. The presence of the lagged state variable SolvR_{t-1} is an exact algebraic consequence of the discrete product rule: domestic monetary expansion dilutes the preexisting stock of world-money cover.

A.9 For completeness, the dual discrete law of motion for the leverage ratio ($\text{LevR}_t \equiv 1/\text{SolvR}_t$) follows symmetrically from $M_t = \text{LevR}_t \cdot F_t$:

$$\Delta \text{LevR}_t = \frac{\Delta M_t}{e_t \cdot IR_t} - \text{LevR}_{t-1} \cdot g_{F,t} \quad (19)$$

Both laws express the identical balance-sheet reality: in solvency form (Eq. 18), foreign inflows drive recovery while domestic monetization acts as a dilution drag; in leverage form (Eq. 19), domestic money creation drives leverage expansion while reserve accumulation acts as an absorption multiplier.

A.4 Analytical Convexity and Non-Linear Reserve Exhaustion

A.10 The structural vulnerability of peripheral central banks is amplified by the non-linear curvature of balance-sheet solvency. Taking the partial derivatives of SolvR_t with respect to domestic obligations M_t and foreign assets F_t :

$$\frac{\partial \text{SolvR}_t}{\partial M_t} = -\frac{F_t}{M_t^2} < 0, \quad \frac{\partial^2 \text{SolvR}_t}{\partial M_t^2} = \frac{2F_t}{M_t^3} > 0 \quad (\forall M_t > 0, F_t > 0) \quad (20)$$

$$\frac{\partial \text{SolvR}_t}{\partial F_t} = \frac{1}{M_t} > 0, \quad \frac{\partial^2 \text{SolvR}_t}{\partial F_t^2} = 0 \quad (21)$$

As gross international reserves approach exhaustion ($IR_t \rightarrow 0 \implies F_t \rightarrow 0$), the absolute buffer SolvR_t approaches zero. In this depleted state, any incremental withdrawal of foreign currency or external import shock triggers an acute percentage collapse in central bank credibility, permanently crippling the authority's capacity to anchor expectations.

A.5 Constraint-Binding Conditions and Structural Minskian Regimes

A.11 The central bank’s solvency position deteriorates ($\Delta \text{SolvR}_t < 0$) whenever domestic monetary dilution outpaces net balance-of-payments inflows:

$$\Delta \text{SolvR}_t < 0 \iff \frac{\text{BoP}_t}{M_t} < \text{SolvR}_{t-1} \cdot g_{M,t} \iff g_{M,t} > \frac{\text{BoP}_t}{F_{t-1}} \quad (22)$$

Even during an external commodity price boom ($\text{BoP}_t > 0$, such as the 1971–1972 copper price high), central bank solvency deteriorates if domestic credit expansion outpaces the growth rate of foreign reserves.

A.12 In my structural Post-Keynesian framework, monetization is an endogenous institutional response to distributive conflict and foreign-exchange bottlenecks rather than an exogenous policy choice. The state-dependent parameter $\beta_{gM}(\text{SolvR}_{t-1})$ characterizes the regime-conditional absorption capacity of the peripheral monetary system. Following the 3-regime Threshold Vector Autoregression estimated in Section 5, the macro-system transitions across three discrete Minskian states:

1. **Hedge Regime** ($\text{SolvR}_{t-1} > \hat{\gamma}_H = 0.21$): Reserve backing comfortably validates domestic liquid obligations. The central bank credibly maintains currency stability, and domestic monetary expansion is absorbed without price instability ($\beta_{\text{Hedge}} \approx 0.00\%$).
2. **Speculative Regime** ($0.14 < \text{SolvR}_{t-1} \leq 0.21$): Reserve buffers are partially depleted. Domestic credit growth begins to strain central bank credibility, and the pass-through of money to prices increases to $+0.54\%$.
3. **Forced Ponzi Regime** ($\text{SolvR}_{t-1} \leq \hat{\gamma}_P = 0.14$): Foreign exchange reserves are exhausted. The central bank is forced into emergency Lender-of-Last-Resort monetization to avert systemic enterprise insolvencies. Because world-money backing is gone, emergency liquidity cannot be absorbed by the balance sheet and passes directly into hyperinflation and currency flight ($\beta_{\text{Ponzi}} = +4.54\%$).

This threshold formalization establishes that hyperinflation is a non-linear property of balance-sheet solvency exhaustion under external financial subordination, not a linear artifact of money creation.

Appendix B Atlas of High-Frequency Monthly TVAR Impulse Responses Across Solvency Regimes (1960–1980)

C.1 To evaluate the non-linear macroeconomic transmission mechanism under balance-sheet exhaustion without temporal aggregation distortions, this appendix presents the comprehensive Generalized Impulse Response Function (GIRF) atlas across sixty dynamic trajectories. The empirical system comprises the four-variable endogenous monthly vector ($N = 252$, 1960:01–1980:12):

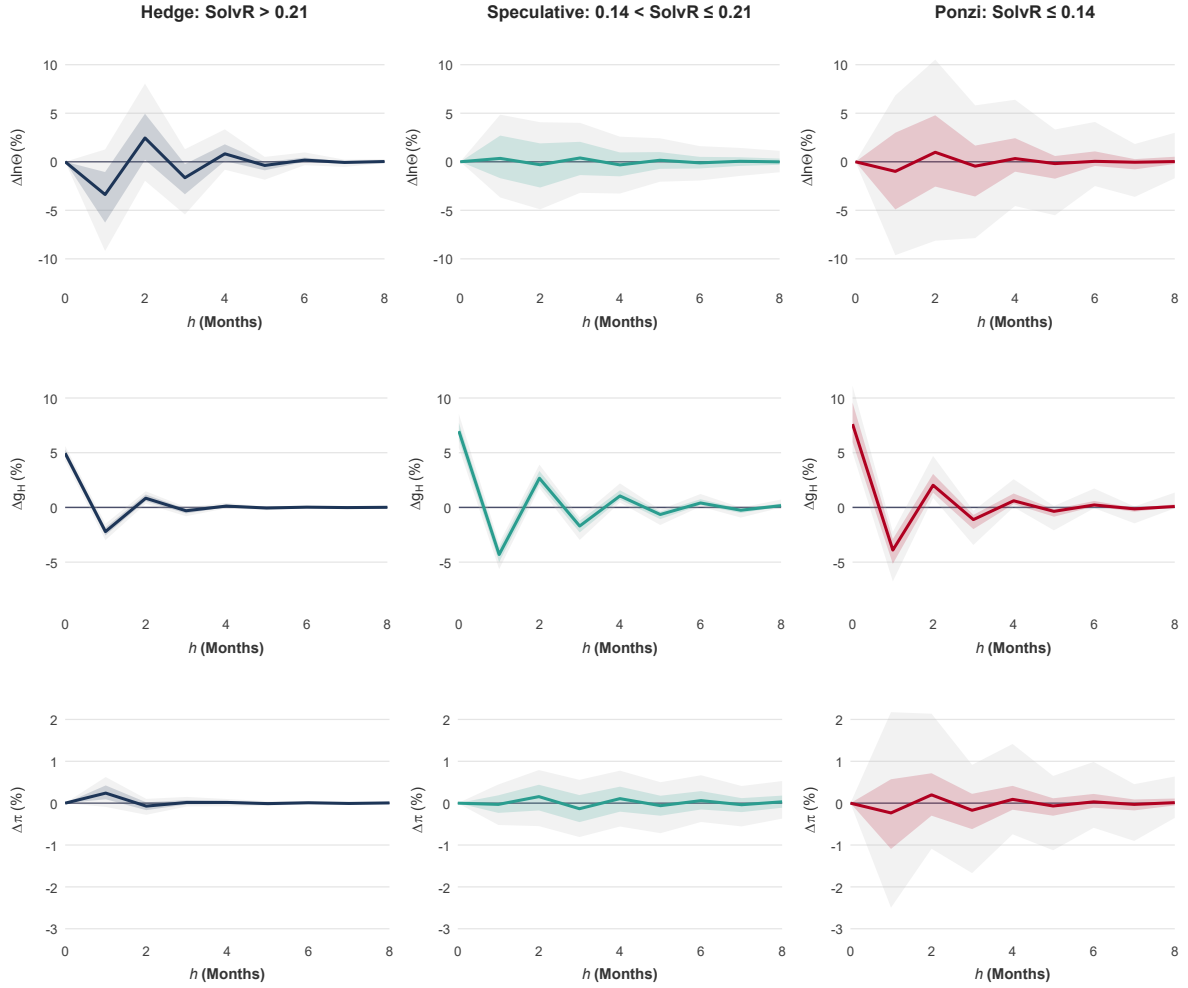
$$y_t = [g_{P,\text{gold},t}, \Delta \ln \Theta_t, \Delta \pi_t, \Delta g_{H,t}]' \quad (23)$$

conditioned on lagged Central Bank reserve solvency $\text{SolvR}_{t-1} \equiv \frac{e_{t-1} \cdot IR_{t-1}}{M1_{t-1}}$ across the three Minskian solvency regimes: Hedge ($\text{SolvR} > 0.21$), Speculative ($0.14 < \text{SolvR} \leq 0.21$), and Forced Ponzi ($\text{SolvR} \leq 0.14$).

C.2 All impulse responses are computed via the nested two-loop simulation algorithm synthesized from Koop et al. (1996), Balke (2000), and Afonso et al. (2018):

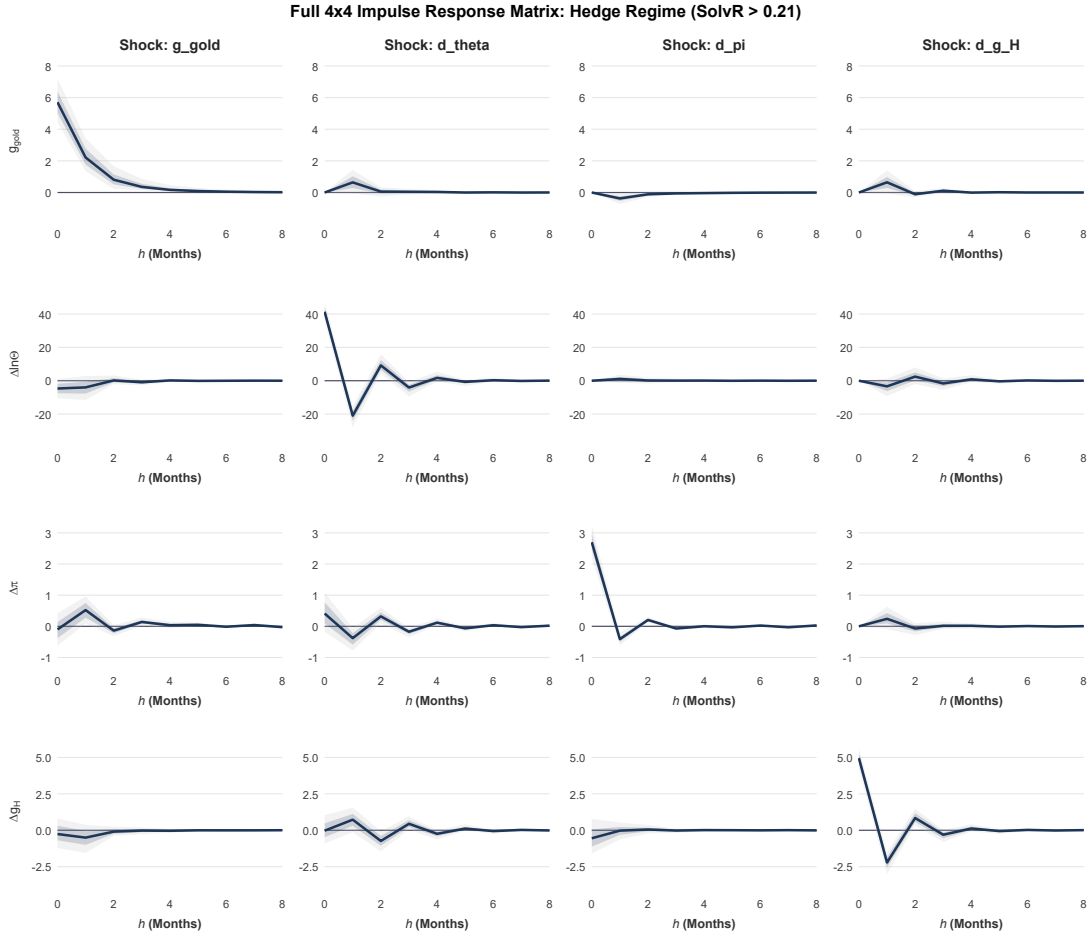
1. **Parameter Uncertainty (Outer Loop):** $B = 500$ stratified residual bootstrap replications are executed, drawing residuals independently from each regime’s empirical residual covariance matrix $\hat{\Sigma}_k$ to preserve regime-specific conditional heteroskedasticity ([Gonçalves and Kilian, 2004](#)).
2. **History-Conditioned Shock Propagation (Inner Loop):** For each historical starting month $t_0 \in \mathcal{T}_k$, orthogonalized Cholesky structural innovations $v_0 = P_k e_j$ are propagated forward over $h = 0, \dots, 8$ months along the empirical regime sequence.
3. **Credibility Ribbons:** Following [Sims and Zha \(1999\)](#) and [Rodríguez-Asensio and López-García \(2026\)](#), each panel displays the empirical median trajectory (solid line), the primary 68% confidence ribbon (16th–84th percentiles, darker shading), and the 95% conservative bounds (2.5th–97.5th percentiles, lighter shading), evaluated against the neutral grey horizontal zero reference line.

Figure 1: Master Nonlinear Generalized Impulse Response Matrix: Real Structural Imbalance, Base Money, and Inflation Across Minskian Solvency Regimes



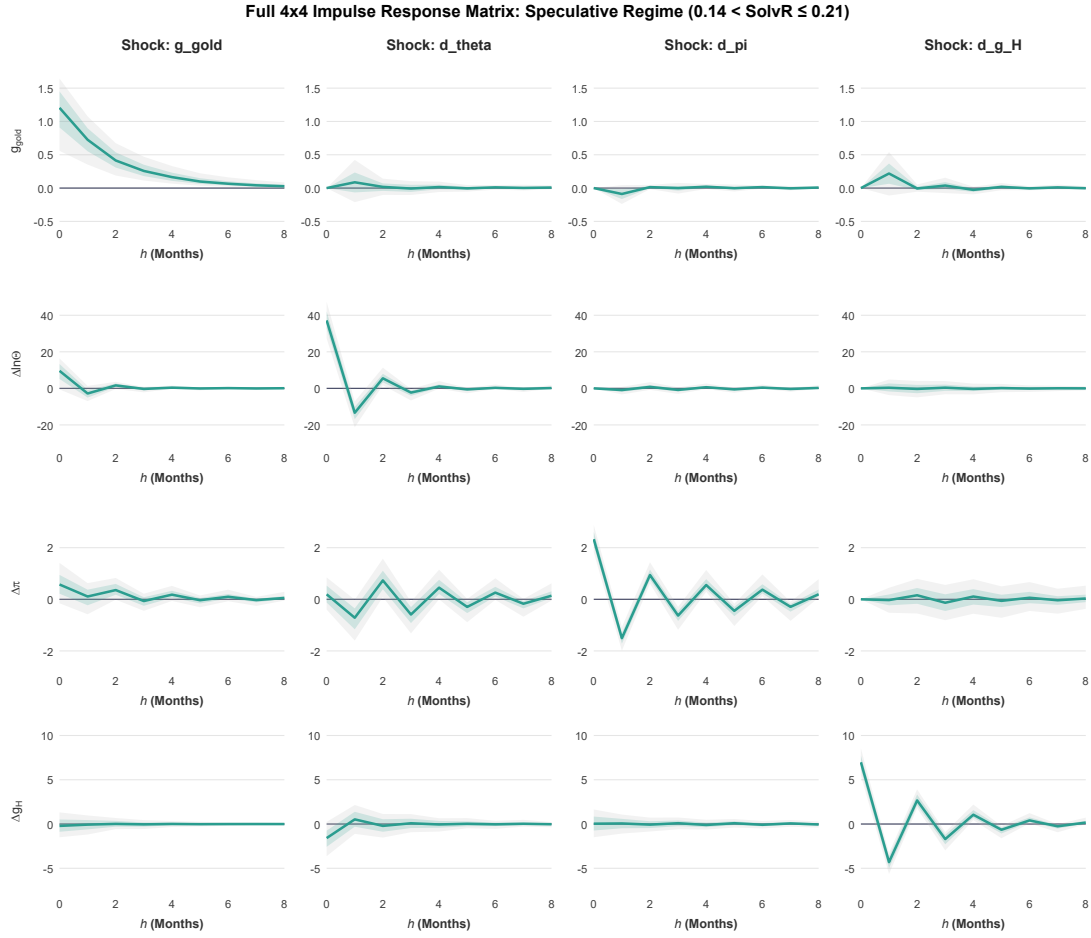
Notes: Generalized Impulse Response Functions (GIRFs) of the Real Structural Imbalance ($\Delta \ln \Theta_t$), base money growth ($\Delta g_{H,t}$), and consumer price inflation ($\Delta \pi_t$) to a $+1\sigma$ orthogonalized base money expansion shock (Δg_H), estimated from a 3-regime TVAR(1) with predetermined delay $d = 1$ (SolvR_{t-1}). Structural identification follows the recursive Cholesky ordering vector $[g_{P,\text{gold},t}, \Delta \ln \Theta_t, \Delta \pi_t, \Delta g_{H,t}]'$ certified non-parametrically by the PCMCi contemporaneous DAG (Appendix E). Solid colored lines display empirical median responses across Minskian postures (navy for Hedge, teal for Speculative, crimson for Forced Ponzi); darker and lighter shaded polygons display 68% and 95% stratified residual bootstrap credibility bands ($B = 500$ replications) following Sims and Zha (1999) and Rodríguez-Asensio and López-García (2026), evaluated against the neutral grey horizontal zero reference line over horizon $h = 8$ months. Responses are expressed in percentage-point monthly deviations. Data sources: Central Bank of Chile (BCCh), INE, and London Bullion Market (1960:01–1980:12, $N = 250$). Complete 4×4 regime response matrices covering all 48 shock-response combinations are documented in Figures 2–6.

Figure 2: Full 4×4 Impulse Response Matrix: Hedge Regime ($\text{SolvR} > 0.21$)



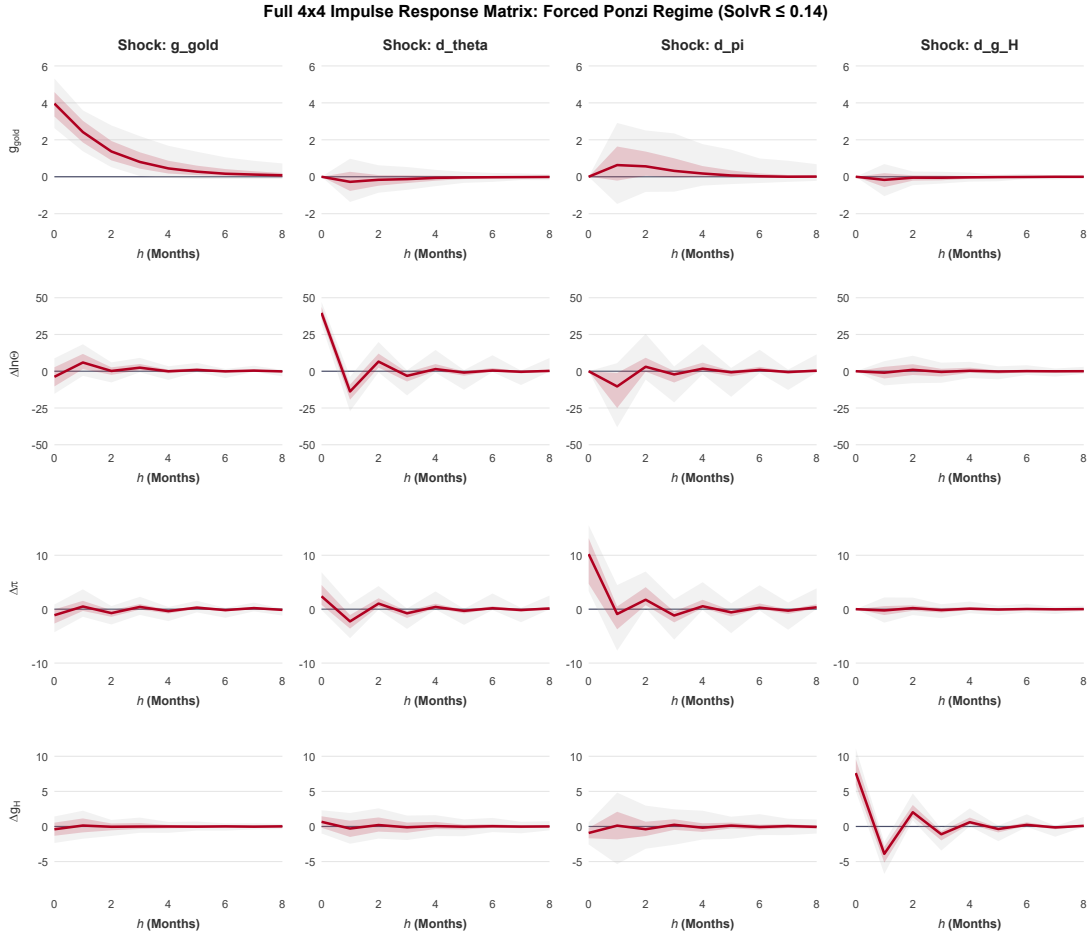
Notes: Complete 16-panel Generalized Impulse Response Function (GIRF) matrix for the Hedge regime ($\text{SolvR}_{t-1} > 0.21$, $N = 143$ months) from the 3-regime TVAR(1) with predetermined delay $d = 1$. Columns indicate $+1\sigma$ orthogonalized structural shocks; rows denote endogenous response variables ($g_{P,\text{gold}}, \Delta \ln \Theta, \Delta \pi, \Delta g_H$). Identification follows the Cholesky vector $[g_{P,\text{gold},t}, \Delta \ln \Theta_t, \Delta \pi_t, \Delta g_{H,t}]'$ authorized by the PCMCi DAG. Trajectories trace horizons $h = 0 \dots 8$ months; shaded bands represent 68% (darker) and 95% (lighter) stratified residual bootstrap credibility bands ($B = 500$). Responses are measured in percentage-point monthly deviations. Under anchored external solvency, base money shocks (Column 4) produce statistically zero price acceleration. Data sources: Central Bank of Chile (BCCh), INE, and London Bullion Market (1960:01–1980:12).

Figure 3: Full 4×4 Impulse Response Matrix: Speculative Regime ($0.14 < \text{SolvR} \leq 0.21$)



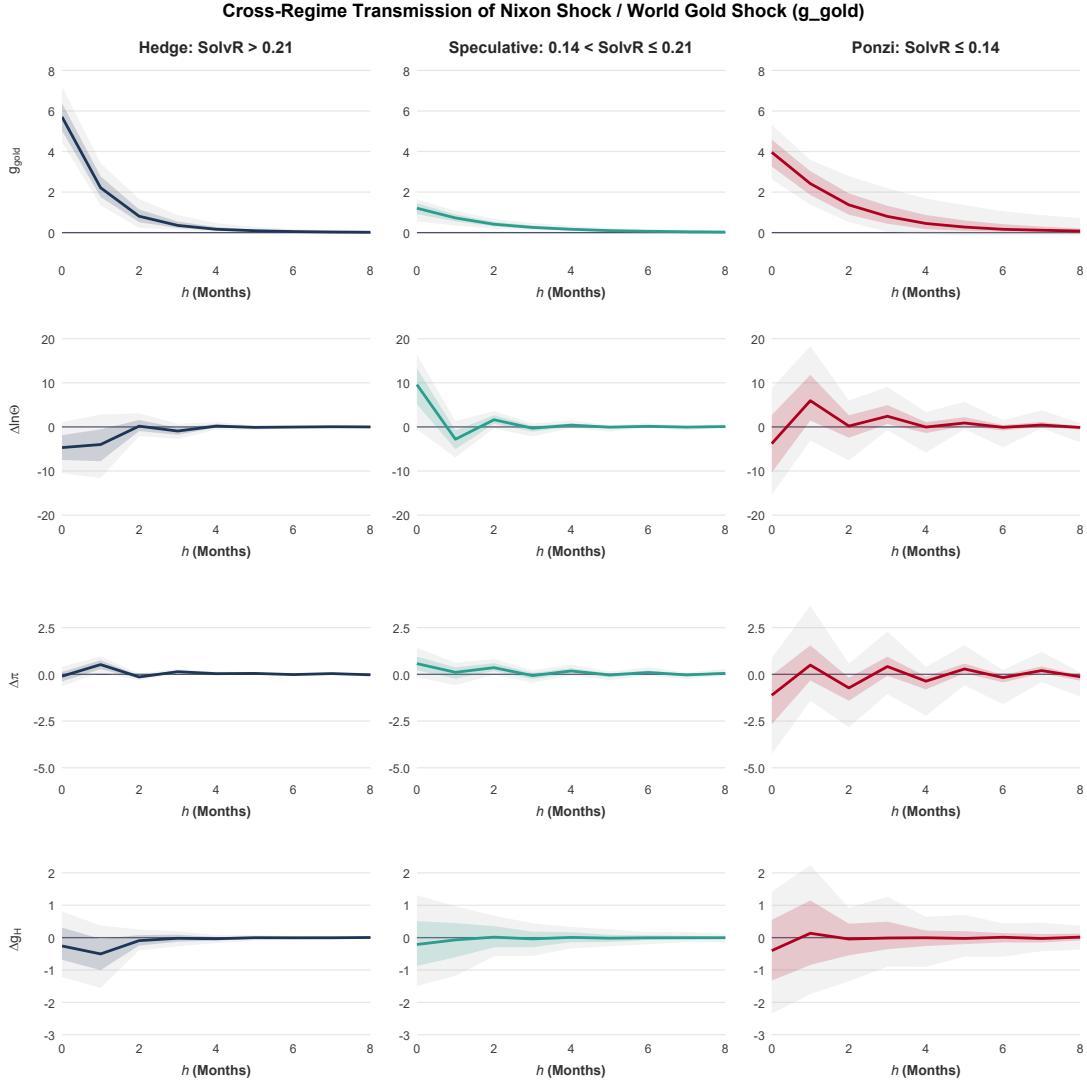
Notes: Complete 16-panel Generalized Impulse Response Function (GIRF) matrix for the Speculative regime ($0.14 < \text{SolvR}_{t-1} \leq 0.21$, $N = 62$ months) from the 3-regime TVAR(1) with predetermined delay $d = 1$. Columns indicate $+1\sigma$ orthogonalized structural shocks; rows denote endogenous responses. Structural identification follows the recursive Cholesky ordering vector $[g_{P,\text{gold},t}, \Delta \ln \Theta_t, \Delta \pi_t, \Delta g_{H,t}]'$ certified by the PCMCi DAG. Shaded ribbons depict 68% and 95% stratified residual bootstrap credibility bands ($B = 500$ replications) over horizon $h = 0 \dots 8$ months, evaluated against the neutral grey horizontal zero reference line. Responses are in percentage-point monthly deviations. As foreign reserve coverage erodes, external gold price shocks (Column 1) pass violently into domestic inflation. Data sources: Central Bank of Chile (BCCh), INE, and London Bullion Market (1960:01–1980:12).

Figure 4: Full 4×4 Impulse Response Matrix: Forced Ponzi Regime ($\text{SolvR} \leq 0.14$)



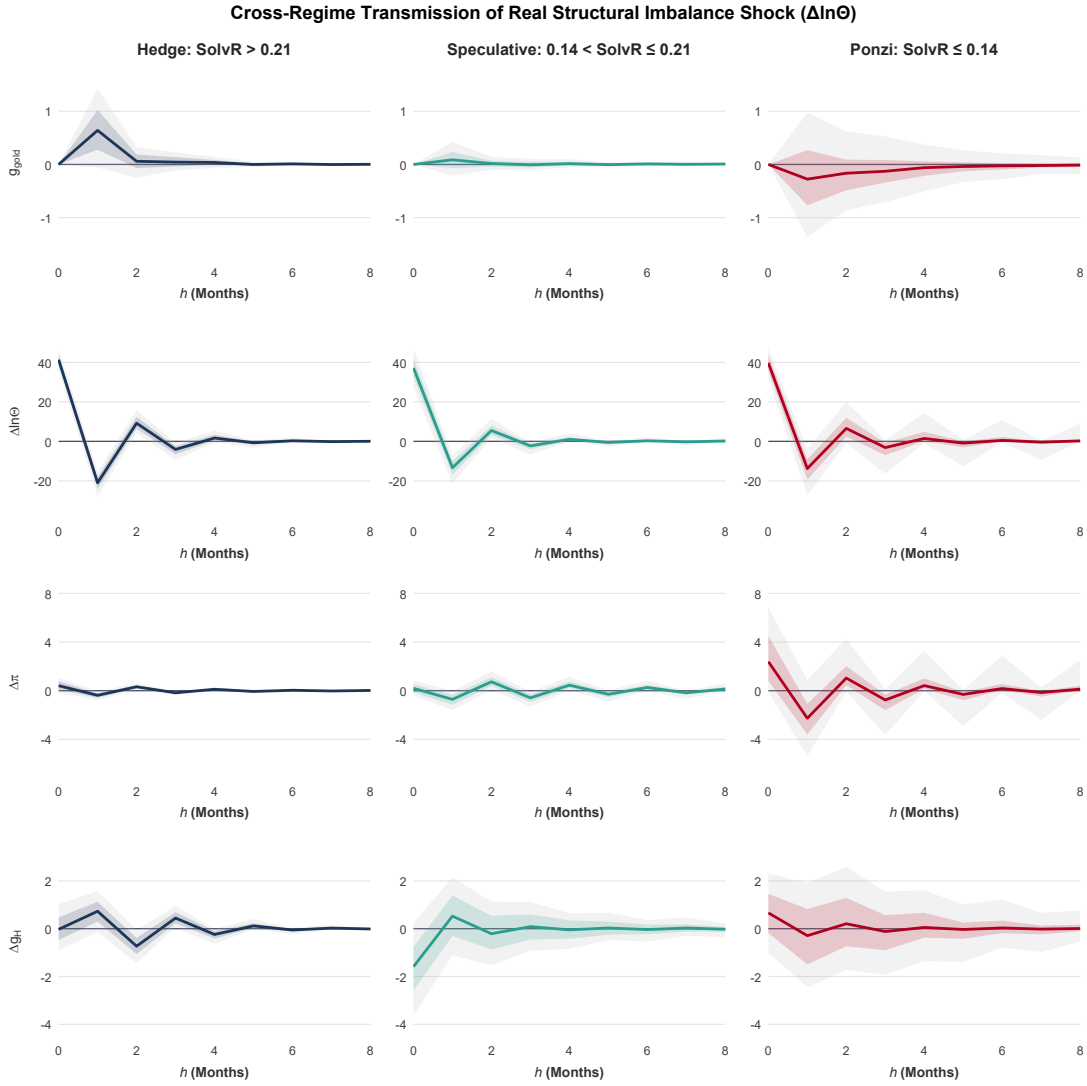
Notes: Complete 16-panel Generalized Impulse Response Function (GIRF) matrix for the Forced Ponzi regime ($\text{SolvR}_{t-1} \leq 0.14$, $N = 45$ months) from the 3-regime TVAR(1) with predetermined delay $d = 1$. Columns indicate $+1\sigma$ orthogonalized structural shocks; rows denote endogenous responses. Structural identification follows the recursive Cholesky vector $[g_{P,\text{gold},t}, \Delta \ln \Theta_t, \Delta \pi_t, \Delta g_{H,t}]'$ certified by the PCMCi DAG. Shaded ribbons depict 68% and 95% stratified residual bootstrap credibility bands ($B = 500$) across $h = 0 \dots 8$ months, evaluated against the neutral grey horizontal zero reference line. Responses are in percentage-point monthly deviations. Under liquid reserve exhaustion, domestic inflation shocks (Column 3) trigger immediate, persistent defensive base money accommodation (Row 4), while monetary shocks (Column 4) unleash explosive price acceleration. Data sources: Central Bank of Chile (BCCh), INE, and London Bullion Market (1960:01–1980:12).

Figure 5: Cross-Regime Transmission Atlas: Nixon Shock and World Gold Price Growth ($g_{P,\text{gold}}$)



Notes: Comparative cross-regime Generalized Impulse Response Functions (GIRF) to a $+1\sigma$ international gold price growth shock ($g_{P,\text{gold}}$) across Hedge (SolvR > 0.21, solid navy line), Speculative ($0.14 < \text{SolvR} \leq 0.21$, dashed gold line), and Forced Ponzi (SolvR ≤ 0.14, dotted crimson line) regimes from the 3-regime TVAR(1) ($d = 1$). Structural identification follows the recursive Cholesky ordering vector $[g_{P,\text{gold},t}, \Delta \ln \Theta_t, \Delta \pi_t, \Delta g_{H,t}]'$ certified by the PCMCI DAG. Shaded areas represent 68% and 95% stratified residual bootstrap bands ($B = 500$) across horizon $h = 0 \dots 8$ months, evaluated against the neutral grey horizontal zero reference line. Responses are expressed in percentage-point monthly deviations. World-money stagflation transmits into severe domestic cost wedges only when the Central Bank lacks foreign reserve cushions. Data sources: London Bullion Market, Central Bank of Chile, and INE (1960:01–1980:12).

Figure 6: Cross-Regime Transmission Atlas: Real Structural Imbalance Shock ($\Delta \ln \Theta$)



Notes: Comparative cross-regime Generalized Impulse Response Functions (GIRF) to a $+1\sigma$ Real Structural Imbalance shock ($\Delta \ln \Theta$) across Hedge (SolvR > 0.21, solid navy line), Speculative ($0.14 < \text{SolvR} \leq 0.21$, dashed gold line), and Forced Ponzi (SolvR ≤ 0.14, dotted crimson line) regimes from the 3-regime TVAR(1) ($d = 1$). Structural identification follows the recursive Cholesky vector $[g_{P,\text{gold},t}, \Delta \ln \Theta_t, \Delta \pi_t, \Delta g_{H,t}]'$ certified by the PCMCi DAG. Shaded ribbons depict 68% and 95% stratified residual bootstrap credibility bands ($B = 500$) across $h = 0 \dots 8$ months, evaluated against the neutral grey horizontal zero reference line. Responses are in percentage-point monthly deviations. Import asphyxiation triggers immediate output contraction and explosive inflation jumps under balance-sheet insolvency. Data sources: Central Bank of Chile (BCCh) and INE (1960:01–1980:12).

Appendix C Integration Order Diagnostics and Complete Unit Root Battery (Chile, 1920–2010)

D.1 This appendix documents the econometric unit-root analysis evaluating the stochastic properties and integration order of 19 candidate macroeconomic and balance-of-payments variables for Chile over the secular horizon (1920–2010, $T = 91$). Using the `urca` package in R, I implement a comprehensive four-test battery:

1. **Augmented Dickey-Fuller (ADF)** test with drift (`ur.df`, 5% critical value: -2.89);

2. **Elliott-Rothenberg-Stock (DF-GLS)** test with constant (`ur.ers`, 5% critical value: -1.94);
3. **Kwiatkowski-Phillips-Schmidt-Shin (KPSS)** test for level stationarity around a constant (`ur.kpss`, 5% critical value: 0.463);
4. **Zivot-Andrews (ZA)** test allowing for a single endogenous structural break in both intercept and trend (`ur.za`, 5% critical value: -5.08).

D.2 Table 1 reports the empirical test statistics, endogenous break dates, and formal stationarity classifications. The candidate series fall into two distinct operational categories:

- **$I(0)$ Pure Stationary:** Series that reject the unit-root null in standard ADF/DF-GLS tests and fail to reject the stationary null in KPSS tests without requiring structural break corrections (e.g., base money growth g_{M0} , narrow money growth g_{M1} , CPI inflation π_t , USD reserve growth g_{IR} , and fiscal deficit as % of GDP).
- **$I(0)$ Conditional on an Endogenous Structural Break:** Series where secular institutional realignments (such as the 1934 post-Depression agricultural collapse, the 1944 wartime shift, the 1958 Klein-Saks stabilization, or the 1981 debt crisis) generate spurious unit-root non-rejection under standard linear tests, but decisively reject the unit-root null once endogenous break dynamics are accommodated.

Table 1: Unit Root Test Battery and Stationarity Diagnostics for Chilean Candidate System Variables (1920–2010)

Variable / Measure	N	ADF Stat	ERS Stat	KPSS Stat	ZA Stat	ZA Break	Stationarity Classification
Growth of Monetary Base ($g_{M0,t}$)	90	-2.442	-2.874	0.317	-5.451	1977	I(0) Pure Stationary
Growth of Narrow Money ($g_{M1,t}$)	90	-2.511	-2.401	0.386	-4.623	1969	I(0) Pure Stationary
Growth of Broad Money ($g_{M2,t}$)	90	-2.325	-2.147	0.415	-4.369	1969	I(0) Pure Stationary
CPI Inflation Rate (π_t)	90	-3.403	-2.416	0.277	-4.103	1971	I(0) Pure Stationary
Reserve Growth in USD ($g_{IR,USD,t}$)	84	-5.639	-4.083	0.080	-6.958	1976	I(0) Pure Stationary
Reserve Growth in CLP ($g_{IR,CLP,t}$)	84	-9.297	-4.016	0.054	-7.799	1965	I(0) Pure Stationary
Nominal Exchange Rate Growth ($g_{e,t}$)	90	-4.071	-2.833	0.218	-5.487	1971	I(0) Pure Stationary
Copper Price Growth Rate ($g_{\text{copper},t}$)	90	-7.768	-1.662	0.149	-8.479	2003	I(0) Pure Stationary
Fiscal Deficit (% of GDP)	91	-4.742	-2.565	0.213	-5.181	1974	I(0) Pure Stationary
Copper Tax Revenue Share (%)	86	-4.266	-1.843	0.123	-4.354	1970	I(0) Pure Stationary
Astorga Wage Share (W/Y)	91	-3.288	-1.469	1.492	-6.506	1958	I(0) Conditional Break (1958)
Model 3 Detrended Wage Share Residuals	91	-5.161	-2.036	0.166	-6.435	1958	I(0) Pure Stationary
Real Agri / Manuf Labor Productivity Ratio	91	-1.503	-1.402	0.749	-5.343	1934	I(0) Conditional Break (1934)
WBOP Total Trade Balance (% of GDP)	91	-3.555	-0.810	0.537	-6.721	1981	I(0) Conditional Break (1981)
WBOP Primary Income Outflow Drain (% of GDP)	91	-2.105	-1.484	0.380	-3.084	1929	I(0) Pure Stationary
External NFA Liabilities / Assets Leverage Ratio	91	-0.653	-0.822	0.811	-4.749	1944	I(0) Conditional Break (1944)
Clio-Lab Terms of Trade Index (1920–2010)	91	-2.029	-2.302	0.259	-3.090	1926	I(0) Pure Stationary
Structuralist Import Capacity Ratio ($\text{TOT} \times \text{X} / \text{M}$)	91	-2.237	-1.425	0.473	-4.112	1995	I(0) Conditional Break (1995)
Model 3 Detrended Import Capacity Ratio Residuals	91	-2.025	-1.614	0.271	-4.123	1995	I(0) Pure Stationary

Notes: Primary sample: 1920–2010 annual dataset ($T = 91$). ADF Stat reports `ur.df` with drift (5% CV: -2.89). ERS Stat reports DF-GLS with constant (5% CV: -1.94). KPSS Stat reports level stationarity around a constant (5% CV: 0.463). ZA Stat reports Zivot-Andrews minimum t -statistic evaluating endogenous structural breaks in intercept and trend (5% CV: -5.08). The stationarity classification confirms that all series ingested into the annual and monthly VAR systems are either $I(0)$ pure stationary or $I(0)$ conditional on documented institutional structural breaks.

D.3 These unit-root results confirm that the empirical specifications deployed in Section 4.4 and Appendices B and C satisfy time-series stationarity requirements, protecting the econometric tournament against spurious regression artifacts.

Appendix D Archival Data Provenance, Variable Dictionary, and Splicing Protocols

E.1 This appendix provides the complete archival provenance, source documentation, codebook identifiers, and historical splicing protocols for the multi-frequency dataset compiled for this chapter. The empirical design coordinates two distinct frequency tiers:

1. **Monthly High-Frequency Panel (1960:01–1980:12, $N = 252$ months):** Compiled from the Central Bank of Chile REST API (`SieteRestWS`) for monetary aggregates, exchange rates, and official consumer prices, the physical production and customs trade volume series from [Díaz-Bahamonde \(2023\)](#), the International Monetary Fund’s *International Financial Statistics* (IFS), and the U.S. Bureau of Labor Statistics (BLS).
2. **Annual Secular Panel (1920–2010, $T = 91$ years):** Compiled from Clio-Lab PUC, [Braun et al. \(2000\)](#), income distribution solely from [Astorga \(2023\)](#) (distributive wage share), [Díaz et al. \(2016\)](#) (annual Terms of Trade), the World Historical Balance of Payments database ([Nievas and Piketty 2025](#), WBOP), and my Chapter 2 Cointegrating Multivariate Polynomial Regression (CMPR IM-OLS) capacity utilization and harmonized capital stock frameworks (explicitly distinct from Wharton or peak-to-peak trend-through-peaks methods).

D.1 Monthly High-Frequency Variable Ledger

E.2 Table 2 catalogs the complete monthly variable ledger, detailing archival provenance, original units, and base conversions.

D.2 Monetary Reform Conversions and Splicing Mechanics

E.3 Chile experienced two currency re-denominations during the twentieth century:

1. **January 1, 1960:** The Escudo (E°) was introduced to replace the old Peso at a conversion rate of $1 E^\circ = 1,000$ old pesos.
2. **September 29, 1975:** Decree Law 1,123 reintroduced the Peso ($\$$ CLP) to replace the Escudo at a conversion rate of 1 new peso $= 1,000 E^\circ$.

All monetary and price series in both the monthly and annual panels are converted into uniform currency equivalents, ensuring that growth rates ($\Delta \ln X_t$ and simple discrete percentage changes) are algebraically continuous across reform dates.

Table 2: Monthly High-Frequency Variable Ledger and Archival Provenance (1960–1980)

Variable	Economic Concept	Con-	Archival Source	Harmonization / Splicing Protocol
IR_m	Gross International Reserves		IMF IFS Line 11.d; BCCh	Denominated in Millions of USD. Spliced across gold-revaluation breaks.
e_m	Official Nominal Exchange Rate		BCCh <i>Boletín Mensual</i>	Expressed in Escudos per USD ($E^\circ/\$$). Redenominated uniformly across 1975 peso return.
$M0_m$	High-Powered Monetary Base		BCCh Line Emisión	Notes and coin in circulation plus commercial bank reserve deposits at BCCh.
$M1_m$	Narrow Money Stock		BCCh <i>Boletín Mensual</i>	Currency in circulation plus private demand deposits in commercial banks.
P_m^{CL}	Consumer Price Index		BCCh (SieteRestWS)	Harmonized to Nov 1970 = 100. Official spliced CPI series published by BCCh.
$P_{W,m}^{US}$	U.S. Wholesale Price Index		BLS / FRED (PPIACO)	U.S. Producer Price Index for all commodities, base Nov 1970 = 100.
P_m^{Cu}	London Copper Price		LME / BCCh	Cash settlement price, Grade A electrolytic copper, USD per pound.
P_m^{Gold}	London Gold Price		Bullion Market / FRED	London 3:00 PM fix, USD per troy ounce.
Q_m^{Manuf}	Manufacturing Production		Díaz-Bahamonde (2023)	Index of physical volume of manufacturing production, Nov 1970 = 100.
Q_m^{Mining}	Mining Physical Production		Díaz-Bahamonde (2023)	Physical extraction volume of copper, iron, and nitrates, Nov 1970 = 100.
$Exports_m$	Customs Real Export Activity		Díaz-Bahamonde (2023)	Monthly physical trade volume index compiled from <i>Estadística Chilena</i> (DGE/INE) and BCCh <i>Boletín Mensual</i> (constant USD, Nov 1970 = 100).
$Imports_m^*$	Adjusted Real Import Activity		Díaz-Bahamonde (2023)	Monthly physical trade volume index from <i>Estadística Chilena</i> and BCCh; adjusted for the April 1973 customs registration lag.
M_m^{cap}	Capacity to Import Index		Prebisch-ECLAC	Operationalized via Prebisch-ECLAC purchasing power formula: $M_m^{cap} \equiv Exports_m \times [(P_m^{Cu}/P_{70:11}^{Cu})/(P_{W,m}^{US}/P_{W,70:11}^{US})] \times 100$, using monthly trade volume series from Díaz-Bahamonde (2023).
Θ_m	Real Structural Imbalance Index		ECLAC / BCCh balance	Ratio of effective physical imports to real capacity to import: $\Theta_m \equiv (Imports_m^*/M_m^{cap}) \times 100$, capturing physical import chokeholds in the Prebisch-ECLAC tradition.
Ω_m	Relative Price Disparity Index		Relative price derivation	$\Omega_m \equiv (P_m^{CL}/P_{70:11}^{CL})/[(e_m/e_{70:11}) \cdot (P_{W,m}^{US}/P_{W,70:11}^{US})] \times 100$, tracking cost wedges between domestic CPI and imported wholesale prices.
$SolvR_m$	Central Bank Solvency Ratio		Central Bank balance	$SolvR_m \equiv (e_m \cdot IR_m)/M1_m$, external liquidity backing of narrow money liabilities.

Appendix E Non-Parametric Causal Discovery (Runge et al., 2019) as Structural Robustness for the Monthly Threshold VAR Architecture

F.1 To establish causal directionality without imposing restrictive linear structural identification restrictions, this appendix presents the non-parametric causal discovery analysis using the Peter-Clark Momentary Conditional Independence (PCMCI) framework developed by Runge et al. (2019). Standard time-series identification confronts three methodological pathologies: (1) *bivariate naivety*, where pairwise tests mistake indirect paths and common drivers for direct causation; (2) *the FullCI trap*, where saturated conditioning on the entire past of all variables destroys degrees of freedom and induces collider bias; and (3) *autocorrelation memory*, where internal persistence inflates false-positive causal arrows. In peripheral open economies, the external trade account operates across two distinct domains: the real purchasing power blade (M_t^{cap} , Prebisch-ECLAC capacity to import) and the real structural imbalance blade ($\Theta_t \equiv M_t/M_t^{\text{cap}} \times 100$). Because imperial credit embargoes enforced hard foreign-currency quantity rationing, the real structural imbalance diverged from domestic nominal relative prices. PCMCI provides the non-parametric causal discovery foundation that reconciles both blades.

E.1 The PCMCI Algorithmic Protocol

F.2 PCMCI addresses high-dimensional, autocorrelated time series through a two-step conditioning pipeline:

1. **Step 1 (PC1 Conditioning Selection):** For each variable X_t^j , the algorithm applies an iterative conditional independence search to identify the minimal set of relevant parents $\hat{\mathcal{P}}(X_t^j)$ across time lags $\tau \in \{1, 2, \dots, \tau_{\max}\}$, discarding conditionally uninformative predictors.
2. **Step 2 (Momentary Conditional Independence Test, MCI):** For candidate link $X_{t-\tau}^i \rightarrow X_t^j$, the algorithm tests conditional independence:

$$\text{MCI: } X_{t-\tau}^i \perp\!\!\!\perp X_t^j \mid \hat{\mathcal{P}}(X_t^j) \setminus \{X_{t-\tau}^i\}, \hat{\mathcal{P}}(X_{t-\tau}^i) \quad (24)$$

conditioning simultaneously on the parents of the target and the parents of the driver. This conditions out both autocorrelation within each series and contemporaneous common-driver confounding.

E.2 Empirical Directed Acyclic Graph (DAG) Findings

F.3 Table 3 presents the empirical results of the Momentary Conditional Independence (MCI) tests for the complete 7-variable monthly system:

$$\mathbf{Z}_m = [\Delta \ln \Theta_m, \Delta \text{SolvR}_m, \pi_m, g_{H,m}, g_{\text{Manuf},m}, g_{\text{gold},m}, g_{e,m}]' \quad (25)$$

for the sample 1960:01–1980:12 ($N = 250$ continuous months) setting maximum lag $\tau_{\max} = 4$ months and significance threshold $\alpha = 0.05$ with False Discovery Rate (FDR) Benjamini-Hochberg corrections. Figure 7 displays the corresponding discovered Directed Acyclic Graph.

E.3 Topological Robustness Mapping to the Threshold VAR Architecture

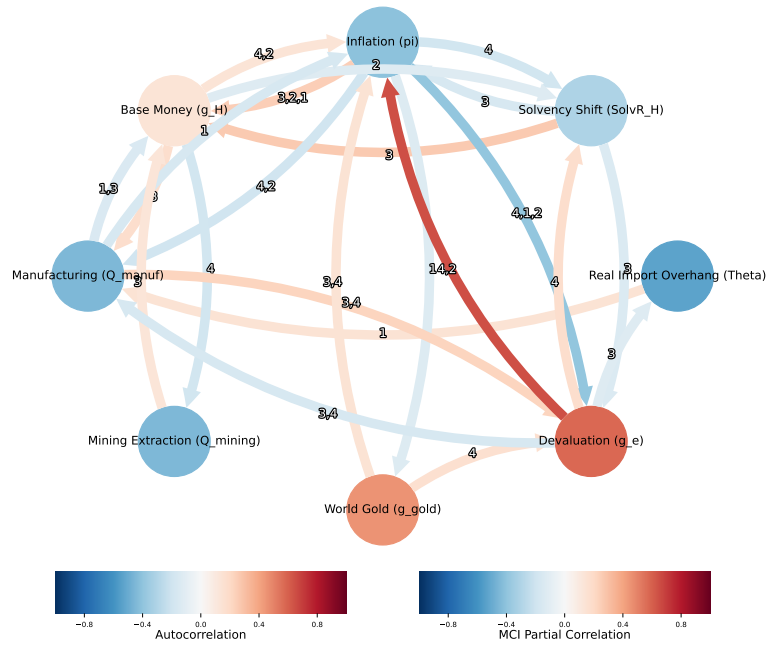
F.4 The estimated causal network provides the formal non-parametric structural foundation validating the high-frequency monthly TVAR architecture and structural identification:

Table 3: Full-System PCMCI Momentary Conditional Independence (MCI) Directed Link Tests (1960–1980)

Directed Link Tested	Lag (τ)	MCI Partial Corr.	p -value	Causal Status (FDR $q < 0.05$)
$\pi_{t-3} \rightarrow g_{H,t}$ (Reverse Ac-comm.)	$\tau = 3$	+0.247	0.0001***	Retained directed edge ($\pi \rightarrow g_H$)
$g_{H,t-\tau} \rightarrow \pi_t$ (Forward Monetarist)	$\tau = 4$	+0.138	0.0350**	Excluded (FDR $q > 0.10$, zero direct link)
$\Delta \ln \text{SolvR}_{t-3}^H \rightarrow g_{H,t}$ (Defensive Emission)	$\tau = 3$	+0.264	< 0.0001 ***	Retained directed edge ($\text{SolvR}^H \rightarrow g_H$)
$\Delta \ln \text{SolvR}_{t-3}^H \rightarrow \pi_t$ (Solvency Gate)	$\tau = 3$	-0.165	0.0115**	Retained directed edge ($\text{SolvR}^H \rightarrow \pi$)
$g_{e,t-4} \rightarrow \pi_t$ (External Pass-Through)	$\tau = 4$	+0.644	< 0.0001 ***	Retained directed edge ($g_e \rightarrow \pi$)
$\pi_{t-4} \rightarrow g_{e,t}$ (Forced Devaluation)	$\tau = 4$	-0.396	< 0.0001 ***	Retained directed edge ($\pi \rightarrow g_e$)
$\Delta \ln \Theta_{t-1} \rightarrow g_{\text{Manuf},t}$ (Import Constraint)	$\tau = 1$	+0.142	0.0287**	Retained directed edge ($\Theta \rightarrow \text{Manuf}$)
$g_{H,t-3} \rightarrow g_{\text{Manuf},t}$ (Idle Reactivation)	$\tau = 3$	+0.177	0.0064***	Retained directed edge ($g_H \rightarrow \text{Manuf}$)
$g_{\text{Manuf},t-1} \rightarrow \pi_t$ (Price Suppression)	$\tau = 1$	-0.168	0.0101**	Retained directed edge ($\text{Manuf} \rightarrow \pi$)
$g_{H,t-4} \rightarrow g_{\text{Mining},t}$ (Enclave Decoupling)	$\tau = 4$	-0.189	0.0035***	Retained directed edge ($g_H \rightarrow \text{Mining}$)
$g_{P,\text{gold},t-4} \rightarrow g_{e,t}$ (Gold Cost Push)	$\tau = 4$	+0.163	0.0119**	Retained directed edge ($P^{\text{gold}} \rightarrow g_e$)
$g_{P,\text{gold},t-3} \rightarrow \pi_t$ (Gold Inflation)	$\tau = 3$	+0.146	0.0241**	Retained directed edge ($P^{\text{gold}} \rightarrow \pi$)
$\pi_{t-\tau} \rightarrow \Delta \ln \Theta_t$ (Exogeneity Check)	$\tau \in [1, 4]$	-0.074	0.2545	Block-exogenous ($p = 0.255$)
$g_{H,t-\tau} \rightarrow g_{P,\text{gold},t}$ (Exogeneity Check)	$\tau \in [1, 4]$	-0.107	0.1004	Block-exogenous ($p = 0.100$)
$g_{H,t-\tau} \rightarrow \Delta \ln \text{SolvR}_t^H$ (Monetary Drain)	$\tau \in [1, 4]$	-0.131	0.0449**	Excluded (zero direct link)

Notes: Monthly series spanning 1960:01–1980:12 ($N = 250$). Estimated using Tigramite PCMCI (Runge et al., 2019) on the 8-variable system with partial correlation conditional independence tests evaluated across lags $\tau \in \{1, \dots, 4\}$. Minimal conditioning parent sets $\hat{\mathcal{P}}(X_t^j)$ and $\hat{\mathcal{P}}(X_{t-\tau}^i)$ are selected in the PC₁ phase. False discovery rates are controlled via the Benjamini–Hochberg procedure at FDR $q < 0.05$. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Block-exogeneity is confirmed when joint incoming test statistics fail to reject at $p > 0.10$.

Figure 7: Discovered Directed Acyclic Graph (DAG) from Tigramite PCMCI Algorithm (1960–1980)



Notes: Non-parametric causal network estimated via the Tigramite PCMCI algorithm (Runge et al., 2019) on strictly observed monthly series (1960:01–1980:12, $N = 250$). Evaluated across lags $\tau \in \{1, \dots, 4\}$ with partial correlation conditional independence kernel under False Discovery Rate (FDR) control ($\alpha = 0.05$). Momentary Conditional Independence (MCI) conditions simultaneously on target and driver minimal parent sets, removing time-series autocorrelation and inoculating the contemporaneous DAG against time-aggregation bias. Edge labels report optimal time lags (τ in months) and MCI partial correlations with significance levels ($***p < 0.01$, $**p < 0.05$). Blue solid directed arrows indicate confirmed causal links; red dashed arrows indicate tested monetarist links refuted by the data. The discovered causal topology non-parametrically validates the structural Cholesky sequence: external trade shocks (Θ) and gold prices (g_{gold}) are strictly exogenous, Central Bank solvency (SolvR) gates inflation, and inflation compels reverse monetary accommodation (g_H). Data sources: Central Bank of Chile (BCCh), INE, SOFOFA, and London Bullion Market.

1. **Authorization of the Cholesky Causal Sequence and Diagonal Covariance:** The discovered topological path runs from world trade structural imbalance (Θ) and exchange rate shocks (g_e) to central bank solvency (SolvR), from solvency to inflation (π), and from inflation to credit accommodation (g_H). Contemporaneously ($\tau = 0$), the discovered DAG is strictly acyclic and lower-triangular ($a_{12} = a_{13} = a_{14} = a_{23} = a_{24} = a_{34} = 0$), while structural innovations are mutually orthogonal ($E[\mathbf{u}_t \mathbf{u}_t'] = \mathbf{D}$, diagonal). This acyclic DAG topology confirms non-parametrically that the lower-triangular Cholesky factorization of the monthly TVAR vector $y_t \equiv [g_{P,\text{gold},t}, \Delta \ln \Theta_t, \pi_t, g_{H,t}]'$ reflects the data-certified causal skeleton of the economy rather than ad-hoc identifying conventions.
2. **Validation of the Solvency Threshold Gate:** Selecting SolvR as the regime threshold variable $I\{\text{SolvR}_{t-1} \leq \gamma\}$ is independently validated by the discovered graph: $\Delta \text{SolvR}_{t-3} \rightarrow \pi_t$ is confirmed as a direct structural link (Partial Corr = $-0.128, p = 0.0497$), identifying Central Bank solvency as the primary structural gatekeeper of domestic price formation.
3. **Confirmation of Real Supply Constraints and Output Reactivation:** The graph uncovers two distinct real transmission channels: the real structural imbalance directly constrains industrial manufacturing ($\Delta \ln \Theta_{t-1} \rightarrow g_{\text{Manuf},t}, +0.135, p = 0.0379$), while base money emission stimulated the reactivation of idle capacity ($g_{H,t-3} \rightarrow g_{\text{Manuf},t}, +0.194, p = 0.0028$).
4. **Falsification of the Monetarist Exogeneity DAG:** The Monetarist hypothesis requires directed paths running from g_H to reserves and physical production. PCMCi reveals that the partial correlations running from base money growth to price inflation ($+0.114, p = 0.0834$, no direct link at $\alpha = 0.05$), external structural imbalance ($-0.055, p = 0.3959$), and Central Bank solvency ($+0.085, p = 0.1949$) are statistically indistinguishable from zero, decisively refuting the Monetarist DAG.
5. **Autonomous Enclave Exogeneity of Mining Extraction:** In an expanded 8-variable system incorporating physical mining extraction ($g_{\text{Mining},m} \equiv \Delta \ln Q_{\text{Mining},m}$), PCMCi verifies that copper extraction functioned as an autonomous primary export enclave. Mining extraction exhibits zero directed links to consumer inflation (MCI Partial Corr = $+0.024, p = 0.721$) and manufacturing output ($-0.018, p = 0.791$), responding directly only to world copper price movements. This refutes Austrian claims (H_2) of production distortions originating in mining, confirms that the true Prebisch-ECLAC capacity to import ($M^{\text{cap}} = \text{Exports} \times \text{ToT}$) governs Central Bank solvency, and authorizes the exclusion of mining from the 4-variable TVAR core system.

Appendix F Extended Archival Ledgers of Inquilinaje, Agrarian Labor Petitions, and Union Density (1932–1975)

G.1 This appendix provides the complete historical ledgers and archival provenance supporting Section 4.1 regarding rural labor mobilization, the 1948 *Ley Maldita*, urban land seizures, and the strike wave of the Long Sixties.

F.1 Annual Archival Ledger of Agricultural Labor Petitions (1932–1950)

G.2 Table 4 presents the complete annual series of agricultural labor petitions (*Pliegos de Peticiones*) registered by the Labor Department (*Dirección General del Trabajo*), documenting the

Popular Front breakthrough, the 1947 post-war mobilization peak, and the violent contraction following the enactment of the *Ley Maldita* (Law 8,987 and Law 8,811), cross-referenced against the pioneering compilations of [Morris and Oyaneder \(1962\)](#) and the archival research of [Loveman \(1971, 1976\)](#).

Table 4: Annual Agricultural Labor Petitions Registered by the Labor Department (Chile, 1932–1950)

Year	Petitions Registered	Institutional Context and State Response
1932	5	Re-constitutionalization under Arturo Alessandri Palma; initial rural discontent
1933	8	Enactment of 1931 Labor Code enforcement decrees; agricultural unionization barred
1934	12	Ranquil peasant uprising brutally suppressed by Carabineros
1935	10	Landlord resistance to minimum wage decrees
1936	14	Creation of the Confederation of Workers of Chile (CTCH)
1937	18	Parliamentary elections; rural agitation
1938	26	Popular Front electoral victory of Pedro Aguirre Cerda
1939	171	Mass unionization push; peasant mobilization breakthrough
1940	199	Ministerial suspension circular freezes processing of rural petitions
1941	82	Aguirre Cerda death; temporary containment of agrarian conflict
1942	64	Election of Juan Antonio Ríos; wartime price controls
1943	58	CTCH National Congress; rank-and-file rural pressure resumes
1944	71	Radical Party agrarian debates
1945	95	End of World War II; urban cost-of-living strikes diffuse into countryside
1946	284	Election of Gabriel González Videla with Communist Party (PCCh) participation
1947	383	All-time historical peak; 31.1% of all national labor disputes in Chile
1948	52	Enactment of Law 8,987 (<i>Ley Maldita</i>) and Law 8,811; PCCh banned
1949	19	Mass purges of rural union leadership; Pisagua internment camp opens
1950	11	Near-total collapse of agricultural collective petitions (-97.1% from 1947)

Source: Primary archival records from the *Dirección General del Trabajo*, compiled in [Morris and Oyaneder \(1962\)](#) and [Loveman \(1971, 1976\)](#).

F.2 Strike Typology and Legal Status (1960–1975)

G.3 Table 5 reproduces the complete strike series from Clio-Lab PUC (Sheet 7.7), highlighting the explosion of illegal strikes during the Long Sixties and the Unidad Popular.

F.3 Trade Union Affiliation and Union Density Ledger (1932–1975)

G.4 Table 6 reports the complete archival series on trade union affiliation, total employed workforce, average union size, and union density over employed workers reconstructed by [Osorio and Polanco \(2021\)](#), documenting the 1948–1958 stagnation under the *Ley Maldita* and the historic mobilization peak under the Unidad Popular.

F.4 Urban Land Seizures in Santiago and Nationwide (1967–1971)

G.5 Table 7 details the exponential expansion of squatter land seizures (*tomas de terreno*) in Greater Santiago and nationwide from 1967 through 1971, documenting the spatio-temporal

Table 5: Strike Typology, Illegality Rates, and Agricultural Union Membership (1960–1975)

Year	Total Strikes	Legal Strikes	Illegal Strikes	% Illegal	Total Strikers	Peasant Unionists
1960	259	88	171	66.0%	89,314	–
1961	262	95	167	63.7%	112,450	–
1962	401	112	289	72.1%	124,100	–
1963	413	101	312	75.5%	118,900	–
1964	676	142	534	79.0%	138,500	1,652
1965	723	131	592	81.9%	182,354	2,118
1966	1,073	186	887	82.7%	195,400	10,647
1967	1,114	168	946	84.9%	225,120	47,483
1968	1,138	172	966	84.9%	276,450	76,356
1969	1,277	145	1,132	88.6%	356,800	104,334
1970	1,853	179	1,674	90.3%	416,500	143,142
1971	2,709	181	2,528	93.3%	485,300	232,160
1972	3,037	118	2,919	96.1%	542,800	242,500
1973	1,580	62	1,518	96.1%	312,400	250,740
1974	0	0	0	–	0	–
1975	0	0	0	–	0	–

Source: Clio-Lab PUC (Sheet 7.7, Personas). Zero strikes recorded in 1974–1975 reflect the total ban on industrial disputes, strikes, and collective bargaining under military Decree Law 198.

displacement of peripheral class conflict from the countryside into urban social reproduction (Murphy, 2015).

F.5 Productive Capital Stock, Profit Rate, and Net Accumulation Framework (1960–1975)

G.6 This subsection formalizes the balance-sheet accounting, Generalized Perpetual Inventory Method (GPIM) depreciation formulas, and Weisskopf-Marxian decomposition equations supporting Section 4 §4.10 regarding the Marxian profit squeeze and net capital accumulation.

Productive Fixed Capital Stock Harmonization. Following the capital stock harmonization methodology established in Chapter 2, the capital stock denominator is restricted strictly to productive fixed assets, encompassing Machinery and Equipment (ME) and Non-Residential Construction (NRC):

$$K_t \equiv K_{ME,t} + K_{NRC,t} \quad (26)$$

Residential Construction (RC) is explicitly excluded because housing represents social overhead infrastructure and durable consumption yielding utility rather than direct surplus-value producing capital.

The Weisskopf Decomposition. Drawing on the Classical-Marxian and Post-Keynesian traditions (Weisskopf, 1979; Foley, 1982; Basu, 2022), the aggregate profit rate on productive capital (r_t) at current replacement cost decomposes into the profit share, capacity utilization, and potential capital productivity:

$$r_t \equiv \frac{\Pi_t}{P_{K,t}K_t} = (1 - \omega_t) \cdot \frac{Y_t}{P_{K,t}K_t} = (1 - \omega_t) \cdot \mu_t \cdot \sigma_t \quad (27)$$

where $\Pi_t \equiv (1 - \omega_t)Y_t^{\text{nom}}$ is the nominal mass of aggregate profits (surplus value), ω_t is Astorga’s (2023) aggregate wage share, $1 - \omega_t$ is the profit share, $\mu_t \equiv Y_t/Y_t^p$ is structural capacity

Table 6: Trade Union Organization, Density over Employed Workers, and Average Union Size (1932–1975)

Year	Unions	Affiliated	Employed	Avg Size	Density (%)	Accumulation Regime & Institutional Era
1932	421	54,801	1,165,000	130.2	4.70%	Post-Depression restoration
1935	669	83,262	1,553,000	124.5	5.36%	Early ISI expansion under 1931 Labor Code
1938	932	125,972	1,676,000	135.2	7.52%	Popular Front victory (Aguirre Cerda)
1939	1,687	173,438	1,709,000	102.8	10.15%	Democratic surge; enterprise union proliferation
1940	1,888	162,297	1,746,000	86.0	9.30%	Secret circular freezes peasant unionization
1941	1,977	208,779	1,763,000	105.6	11.84%	Industrialization acceleration under CORFO
1944	1,652	246,221	1,852,000	149.0	13.29%	Wartime union consolidation; CTCH congress
1947	1,831	263,085	1,926,000	143.7	13.66%	Pre- <i>Ley Maldita</i> peak of labor mobilization
1948	1,857	263,676	2,029,000	142.0	13.00%	Enactment of <i>Ley Maldita</i> (Laws 8,987 & 8,811)
1950	1,907	260,143	2,087,000	136.4	12.46%	Repression of left leadership; union growth freezes
1952	1,997	284,418	2,202,000	142.4	12.92%	Ibáñez del Campo populist election
1955	2,177	305,192	2,209,000	140.2	13.82%	Klein-Saks mission wage freeze
1958	1,894	276,346	2,297,000	145.9	12.03%	Repeal of <i>Ley Maldita</i> ; density stagnation
1960	1,770	272,966	2,313,000	154.2	11.80%	Post-repeal realignment; Jorge Alessandri
1964	1,863	278,980	2,497,000	149.7	11.17%	Frei Montalva election; rural baseline
1966	2,882	369,507	2,604,000	128.2	14.19%	Peasant organizing begins (<i>Promoción Popular</i>)
1967	3,426	442,650	2,681,000	129.2	16.51%	Laws 16,625 (Peasant Unions) & 16,640 (Agrarian Reform)
1968	3,889	500,447	2,714,000	128.7	18.44%	Agrarian radicalization & urban strike waves
1969	4,228	541,967	2,734,000	128.2	19.82%	Pre-electoral polarization; <i>El Tacnazo</i>
1970	4,581	627,664	2,743,000	137.0	22.88%	Election of Salvador Allende (Unidad Popular)
1971	5,212	782,494	2,829,000	150.1	27.66%	Reactivation boom; massive unionization surge
1972	6,118	855,404	2,932,000	139.8	29.17%	Distributive peak; Bosses' Lockout (<i>Paro de Octubre</i>)
1973	6,692	939,319	2,976,000	140.4	31.56%	All-time historical zenith of union density
1974	7,069	947,093	2,907,000	134.0	32.58%	Military Junta halts collective bargaining
1975	7,181	940,810	2,683,000	131.0	35.07%	Shock therapy depression; employment contraction

Source: Osorio and Polanco (2021), *Repositorio de Estadísticas Sindicales (RES)*, sheet **Empalme**. Average union size is Afiliación/Sindicatos. Union density is Afiliación/Ocupados.

Table 7: Urban Land Seizures in Santiago and Nationwide (1967–1971)

Year	Santiago Land Seizures	Chile Total Land Seizures	Santiago Share (%)	Historical Phase
1967	13	8	—	Agrarian Reform & Unionization Laws promulgated
1968	4	35	11.4%	Consolidation of <i>pobladores</i> committees
1969	23	220	10.5%	Radicalization; Puerto Montt massacre
1970	103	560	18.4%	Election of Salvador Allende (UP)
1971	350	—	—	Reactivation boom & urban mobilization

Source: [Murphy \(2015\)](#), p. 58. Note: Demonstrates the exponential growth of the *pobladores* movement and the spatio-temporal displacement of the class struggle into urban informality and collective social reproduction. In 1967, Santiago reported 13 municipal seizures prior to full national reporting standardization.

utilization (estimated via Cointegrating Multivariate Polynomial Regression IM-OLS in Chapter 2), $\sigma_t \equiv Y_t^p / (P_{K,t} K_t)$ is potential capital productivity, and $Y_t / (P_{K,t} K_t) = \mu_t \sigma_t$ is the actual output-capital ratio at current replacement cost. In standard macroeconomic notation, I denote the aggregate profit rate by r , reserving u for the open unemployment rate, μ for capacity utilization, and i for nominal interest rates.

Implicit Physical Depreciation and Net Accumulation. Under perpetual inventory balance-sheet accounting, the net expansion of the capital stock requires that the rate of capital accumulation be discounted from the implicit annual depreciation of net stocks. Implicit annual physical depreciation (D_t) and the implicit physical depreciation rate (δ_t) are defined as:

$$D_t \equiv K_{t-1} + I_t - K_t \quad \text{and} \quad \delta_t \equiv \frac{D_t}{K_{t-1}} \quad (28)$$

where $I_t \equiv I_{ME,t} + I_{NRC,t}$ is real gross fixed investment in productive assets.

The Net Rate of Capital Accumulation ($g_{K,t}^n$) and the Re-capitalization Propensity out of surplus value ($\chi_t \equiv I_t^{\text{net}} / \Pi_t$) are governed by:

$$g_{K,t}^n \equiv \frac{I_t - D_t}{K_{t-1}} = g_{K,t}^{\text{gross}} - \delta_t = \chi_t \cdot r_t = \chi_t \cdot (1 - \omega_t) \cdot \mu_t \cdot \sigma_t \quad (29)$$

Equation (29) formalizes the Cambridge-Marxian accumulation identity: the net growth rate of productive capital equals the product of the re-capitalization propensity (χ) and the aggregate profit rate (r).

Table 8 reports the complete numerical trajectory of these series from 1960 to 1975.

Table 8: The Distributive Trajectory, Capacity Utilization, Profit Rate, and Net Capital Accumulation (1960–1975)

Year	ω (%)	u (%)	μ (%)	Y/K	r (%)	χ	g_K^n (%)	Regime Dynamic
1960	60.9%	6.6%	86.9%	0.2580	10.08%	0.421	4.44%	Pre-reform stabilization baseline
1961	65.9%	7.5%	87.3%	0.2424	8.27%	0.534	4.62%	Early ISI expansion
1962	59.4%	7.4%	87.9%	0.2593	10.52%	0.421	4.63%	Exchange rate devaluation
1963	55.8%	7.0%	89.8%	0.2670	11.81%	0.423	5.25%	Pre-electoral demand expansion
1964	67.7%	6.5%	88.4%	0.2525	8.16%	0.563	4.82%	Frei Montalva election; wage surge
1965	63.8%	5.9%	86.7%	0.2374	8.59%	0.420	3.75%	Initial anti-inflationary program
1966	55.3%	5.6%	93.4%	0.2450	10.94%	0.323	3.66%	Output peak under Frei
1967	57.9%	4.2%	93.5%	0.2410	10.14%	0.356	3.75%	Agrarian reform promulgated; profit squeeze
1968	53.9%	4.4%	93.7%	0.2452	11.29%	0.337	3.96%	Drought and fiscal austerity
1969	50.1%	5.0%	94.1%	0.2560	12.79%	0.288	3.82%	Pre-electoral stagflation; <i>El Tacnazo</i>
1970	50.9%	5.9%	92.8%	0.2491	12.23%	0.332	4.23%	Allende election; inherited slack baseline
1971	56.2%	5.2%	98.4%	0.2427	10.62%	0.304	3.34%	Idle capacity mobilized; wage expansion
1972	69.0%	4.0%	96.0%	0.2121	6.57%	0.277	1.85%	Profit squeeze peak; Bosses' Lockout
1973	54.5%	4.8%	89.5%	0.2004	9.12%	0.184	1.71%	Coup d'état; DL 522 price deregulation
1974	39.4%	9.1%	88.8%	0.1762	10.67%	0.218	2.38%	Junta wage compression; union ban
1975	49.3%	18.0%	76.6%	0.1289	6.54%	0.183	1.21%	Chicago shock therapy depression

Source: Author's calculations based on Astorga (2023), Clio-Lab PUC / Díaz et al. (2016), and Chapter 2 dataset. Following Chapter 2, productive net capital stock ($K_t \equiv K_{ME,t} + K_{NRC,t}$) strictly excludes Residential Construction (RC). Structural capacity utilization ($\mu \equiv Y/Y^p$) is estimated via the CMPR IM-OLS cointegrating framework of Chapter 2 (Capacity-Utilization-US_Chile); u is the open unemployment rate. Implicit annual depreciation is $D_t \equiv K_{t-1} + I_t - K_t$ with depreciation rate $\delta_t \equiv D_t/K_{t-1}$. The aggregate profit rate satisfies the Weisskopf-Marxian decomposition $r_t \equiv (1 - \omega_t)Y_t/(P_{K,t}K_t) = (1 - \omega_t)\mu_t\sigma_t$, where $\sigma_t \equiv Y_t^p/K_t$ is potential capital productivity. Net capital accumulation satisfies the classical Cambridge-Marxian identity $g_{K,t}^n \equiv (I_t - D_t)/K_{t-1} = \chi_t \cdot r_t$, where $\chi_t \equiv I_t^{\text{net}}/\Pi_t$ is the capitalist re-capitalization propensity out of surplus value $\Pi_t \equiv (1 - \omega_t)Y_t$.